

Intelligent Skiing Posture Detection and Recognition Through Internet of Bodies

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ABSTRACT

The training of skiing should start from the control of body posture ability to highlight the characteristics of the sports. Thus, the athletes can have the sports ability in the process of high-speed skiing. This paper establishes a system to automatically recognize the skiing posture which can help athletes grasp the skiing postures. First, the skiing images are collected by distributed camera. Second, the skeleton features are extracted to learn a classification model which is used to recognize and adjust skiing postures. Lastly, the analytical results of posture recognition are returned to athletes through internet of bodies. The framework can effectively recognize the skiing postures and provide athletes with training advice.

KEYWORDS

Classification Model, Internet of Bodies, Posture Recognition, Skeleton Features

1. INTRODUCTION

The technique of skiing, especially alpine skiing, is very difficult, which requires athletes to have a high level of special ability to ensure the exertion of the technique. Special ability refers to various abilities which are suitable for the characteristics of the project and can improve the performance in this project. For skiing, the cultivation of special ability should not only enhance the overall physical quality (Ahmed et al. 2021) based on strength, but also improves the skill to control body posture. The ability to control ability body posture (Horak et al. 2002) is a special ability of skiers. The improvement of this ability has an important impact on the competition.

From the development process of skiing projects, we need to constantly explore new skills and strategies, then combine and improve them. With the progress of sports biomechanics and sports training (Ying & Huang 2021, Pellegrini et al. 2018), the method of sports technology analysis has been greatly improved. The research on posture training and correction (Han et al. 2017, Kashuba et al. 2019) is a hot topic in the scientific research of skiing. The distortion of body posture may lead to skiing failure and induce injury (Rentenberger et al. 2021, Yoshioka et al. 2018). When passing through various uneven areas, the speed of athletes who lose their balance due to large distortion of body posture will be significantly slower than the athletes who maintain a stable skiing posture. It is necessary to pay attention on reducing leaping movements and various leaping body movements during skiing. The key is to keep the balance of body.

In the skiing competition, body posture has a direct impact on competition results. Through the repeated the experimental results in training practice, it is proved that the distortion of posture is induced by the compound participation of posture reflex (Yoshioka et al. 2018, Alhammoud

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2020). For example, an adult athlete who wants to improve his sports performance will raise his hand unconsciously, close his eyes, and cause the excitement of anterior tibial muscle related to joint movement. Therefore, it can be said that the distortion of body posture is directly related to the participation of conditioned reflex. This change in body posture itself is the instinctive physiological mechanism to maintain the best movement. However, in the skiing competition, even if it becomes a physiological mechanism to restore balance, unconscious actions will also cause distortion in posture and affect competition results. This contradiction is mainly solved in training through two methods. One is to train athletes to get used to the conditions of physical changes. The other is to improve the ability to predict physical changes which can be improve with the help of automatic posture recognition and correction system.

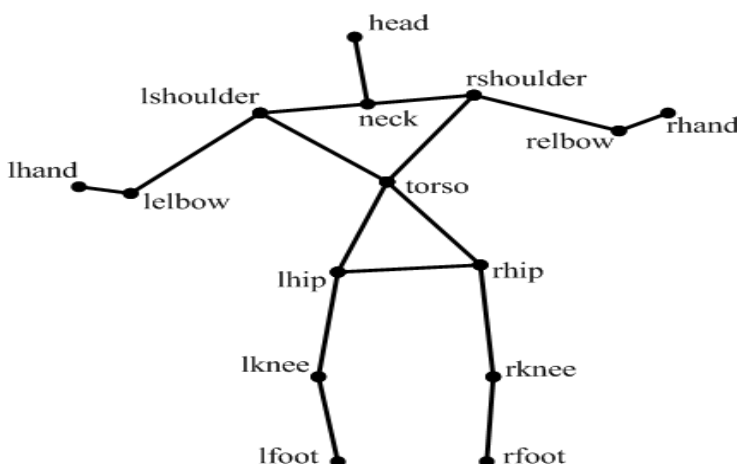
Kinect somatosensory device developed by Microsoft can easily collect human joint point data, scene depth information and color images (Li & Xiao 2018, Wang et al. 2019). It has attracted the attention of many researchers due to its excellent performance. The Kinect has been used in many applications, such as robot navigation (Li & Wang 2020), target tracking (Pang & Liang 2019), object recognition (Yang et al. 2021), and 3D modeling (Cai 2019).

Recently, using Kinect to study human behavior and posture recognition has become a hot spot in the field of behavior recognition (Ma & Guo 2019, Li 2020). In this paper, the Kinect is used to collect joint point data of skiing players. The joint point data is used to analyze the posture of skiing players. According to the characteristics of human structure, human structure vector is constructed by using joint point data as human model in three-dimensional space. The posture features are extracted by calculating the vector angle and vector modulus ratio between human structure vectors. Then, the features of skiing player posture is matched and corrected by the dynamic time warping (DTW) algorithm (Permanasari et al. 2019, Yadav & Alam 2018).

2. SKIING PLAYER JOINT POINTS FROM KINECT SENSOR AND FEATURE REPRESETATION

When an observer stands toward Kinect camera, the positive direction of X-axis points to the right of the observer, the positive direction of Y-axis points to the observer's head, the positive direction of Z-axis is consistent with the observation direction of Kinect camera, and the Z-axis represents depth information. The Kinect camera can collect 15 joint points to represent different parts of the human body. The distribution of each joint point is shown in Figure 1.

Figure 1. The architecture of document type recognition framework



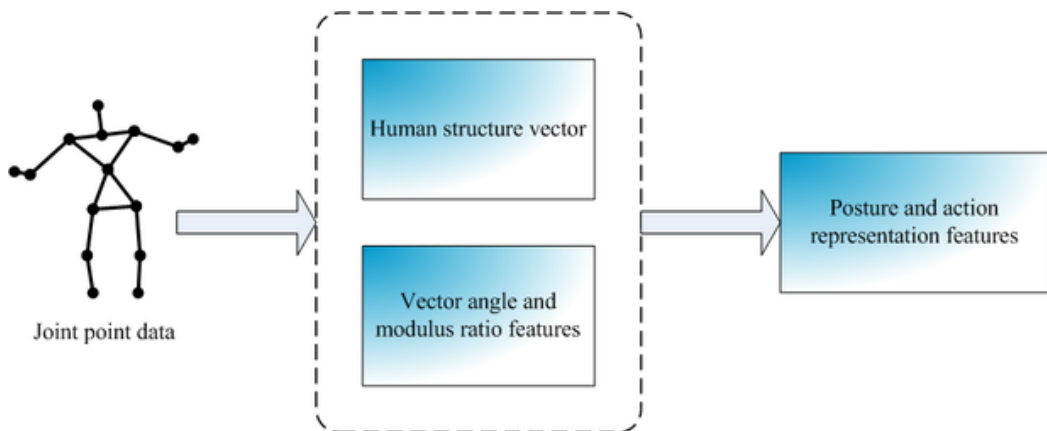
Each joint point is assigned as a name. The details are illustrated in Table 1.

Table 1. The performance comparison of document type recognition

Position	Name	Position	Name	Position	Name
Head	Head	Neck	Neck	Left hand	Lhand
Left elbow	Lelbow	Left shoulder	Lshoulder	Right hand	Rhand
Right elbow	Relbow	Right shoulder	Rshoulder	Center node	Torso
Left hip	Lhip	Left knee	Lknee	Left foot	Lfoot
Right hip	Rhip	Right knee	Rknee	Right foot	rfoot

Due to the difference of human posture and the change of relative position relationship with Kinect, it may increase the difficulty of behavior recognition and classification to directly use joint point information as behavior representation feature. In order to use Kinect joint point data for human behavior analysis, the joint point data should be preprocessed first. Considering human cognitive characteristics and human structure characteristics, this paper first uses joint point data to construct human structure vector to express the posture of human limbs, trunk and head. The human behavior information is mostly contained in these posture information. Based on the structure vector, the features of behavior representation is obtained by calculating the angle and modulus ratio between vectors. The procedure is summarized in the following figure.

Figure 2. The architecture of document type recognition framework



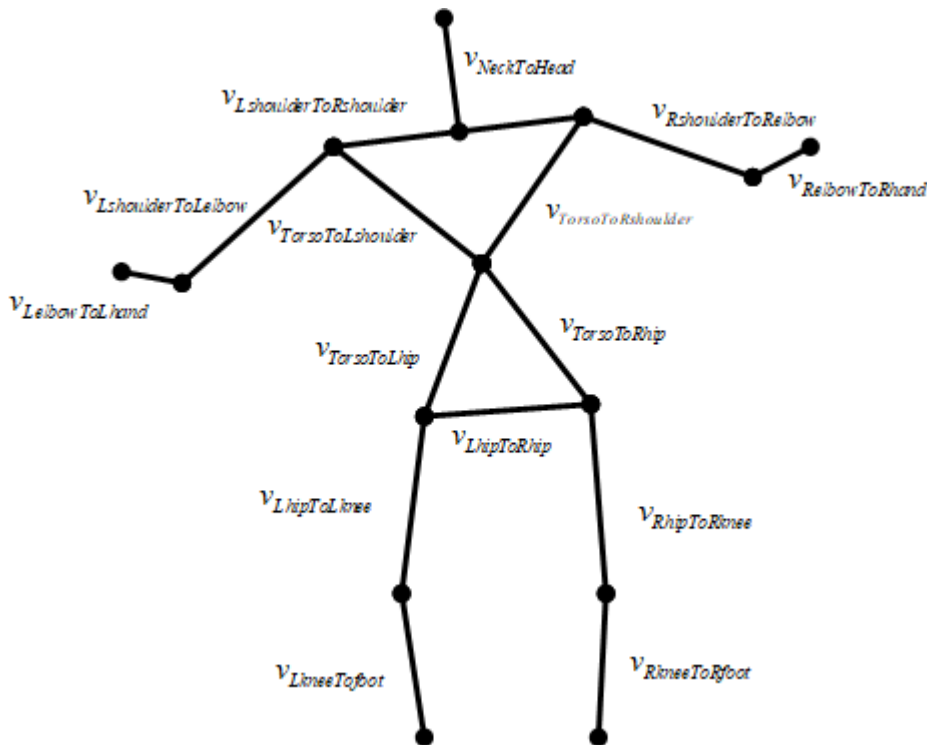
By analyzing human posture, the human structure vector is constructed by selecting appropriate joint point data according to the characteristics of human structure. The human structure vector is very simple. Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two joint points, the corresponding human structure vector is represented as follows:

$$v_{AB} = (x_2 - x_1, y_2 - y_1, z_2 - z_1) \quad (1)$$

According to Equation (1) $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ represent rshoulder and relbow, the vector in Equation (1) describes the posture of the right boom. By combing with the actual situation, the posture may contain more human action information.

In order to describe human actions during skiing, 15 groups of human structure vectors are constructed in this paper. These human structure vectors are divided into three types according to different body parts, including upper limb part, lower limb part, and trunk part. An illustration of human structure vectors is shown in Figure

Figure 3. The architecture of document type recognition framework



The upper limb part includes the human structure vectors from left shoulder to left elbow, from left elbow to left hand, right shoulder to right elbow, from right elbow to right hand. These human structure vectors are denoted as $v_{LshoulderToLeibow}$, $v_{LeibowToLhand}$, $v_{RshoulderToRelbow}$, and $v_{RelbowToRhand}$, respectively. The lower limb part includes the human structure vectors from left hip to left knee, from left knee to left foot, right hip to right knee, from right knee to right foot. These human structure vectors are denoted as $v_{LhipToLknee}$, $v_{LkneeToLfoot}$, $v_{RhipToRknee}$, and $v_{RkneeToRfoot}$, respectively. The trunk part includes the human structure vectors from neck to head, from left shoulder to right shoulder, from torso to left shoulder, from torso to right shoulder, from torso to left hip, from torso to right hip, and from left hip to right hip. These human structure vectors are denoted as $v_{NeckToHead}$, $v_{LshoulderToRshoulder}$, $v_{TorsoToLshoulder}$, $v_{TorsoToRshoulder}$, $v_{TorsoToLhip}$, $v_{TorsoToRhip}$, $v_{LhipToRhip}$ respectively.

When the human body is moving, its posture and the distance to Kinect camera often change, which results in large changes in the defined human body structure vector. However, the actual situation

is that the change trend of limbs during human movement is roughly similar. By selecting the angle between structural vectors, it can effectively eliminate the differences between human structural vectors of different people or different positions. The angle between two vectors

$a = (a_1, a_2, a_3)$ and $b = (b_1, b_2, b_3)$ is written as follows:

$$\theta = \arccos \frac{a \cdot b}{\|a\| \|b\|}, \|a\| \neq 0 \ \& \ \|b\| \neq 0 \quad (2)$$

In the Equation (2), $a \cdot b$, $\|a\|$, and $\|b\|$ are defined as follows:

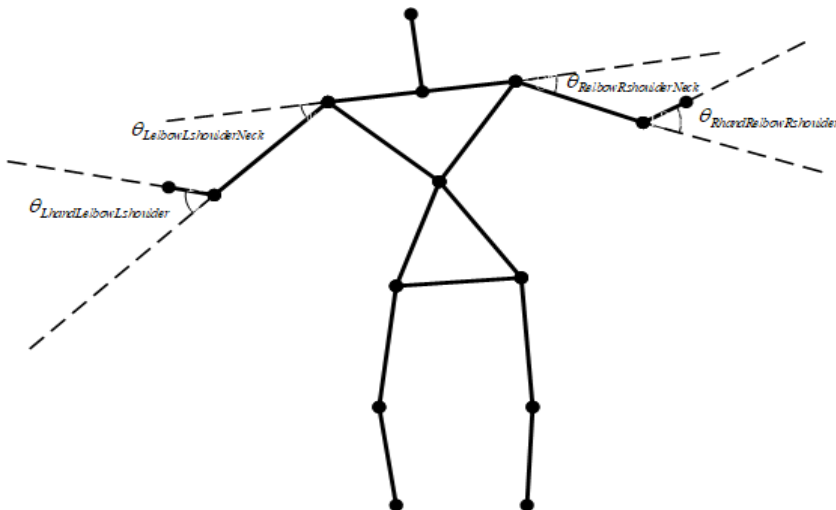
$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3 \quad (3)$$

$$\|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad (4)$$

$$\|b\| = \sqrt{b_1^2 + b_2^2 + b_3^2} \quad (5)$$

According to the human structure vectors, the angles are classified as three parts: upper limb angles, lower limb angles, and intermediate junction angles. The upper limb angles are used to describe the posture or movement of arms. The lower limb angles are used to describe the posture or movement of legs. The angle information at the middle junction is mainly used to represent the changes of human limbs relative to the trunk and the changes of human trunk itself during movement. Figure 4 shows the angle information of upper limbs

Figure 4. The architecture of document type recognition framework



In the Figure 4, the angles of upper limbs include the angle between $v_{L\text{ShoulderToLelbow}}$ and $v_{L\text{ElbowToLhand}}$, the angle between $v_{L\text{ShoulderToLelbow}}$ and $v_{L\text{ShoulderToRshoulder}}$, the angle between $v_{R\text{ShoulderToRelbow}}$ and $v_{RelbowToRhand}$, and the angle between $v_{R\text{ShoulderToRelbow}}$ and $v_{L\text{ShoulderToRshoulder}}$. These angles are denoted as $\theta_{L\text{handLelbowLshoulder}}$, $\theta_{L\text{elbowLshoulderNeck}}$, $\theta_{R\text{handRelbowRshoulder}}$, and $\theta_{R\text{elbowRshoulderNeck}}$.

The lower limb angles include the angle between $v_{L\text{HipToLknee}}$ and $v_{L\text{HipToRhhip}}$, the angle between $v_{R\text{kneeToRfoot}}$ and $v_{R\text{hipToRknee}}$, and the angle between $v_{R\text{hipToRknee}}$ and $v_{L\text{HipToRhhip}}$. These angles are denoted as $\theta_{L\text{hipLkneeLfoot}}$, $\theta_{L\text{kneeLhipRhhip}}$, $\theta_{L\text{hipRhhipRknee}}$, and $\theta_{R\text{hipRkneeRfoot}}$.

The intermediate junction angles include the angle between $v_{\text{HeadToNeck}}$ and $v_{L\text{ShoulderToRshoulder}}$, the angle between $v_{L\text{ShoulderToRshoulder}}$ and $v_{\text{TorsoToLshoulder}}$, the angle between $v_{L\text{ShoulderToRshoulder}}$ and $v_{\text{TorsoToRshoulder}}$, the angle between $v_{\text{TorsoToLshoulder}}$ and $v_{\text{TorsoToRshoulder}}$, the angle between $v_{\text{TorsoToLshoulder}}$ and $v_{\text{TorsoToLhip}}$, the angle between $v_{\text{TorsoToRshoulder}}$ and $v_{\text{TorsoToRhhip}}$, the angle between $v_{\text{TorsoToLhip}}$ and $v_{\text{TorsoToRhhip}}$, the angle between $v_{\text{TorsoToRhhip}}$ and $v_{L\text{HipToRhhip}}$, and the angle between $v_{\text{TorsoToRhhip}}$ and $v_{L\text{HipToRhhip}}$. These angles are denoted as $\theta_{\text{HeadNeckShoulder}}$, $\theta_{R\text{shoulderLshoulderTorso}}$, $\theta_{L\text{shoulderRshoulderTorso}}$, $\theta_{L\text{shoulderTorsoRshoulder}}$, $\theta_{\text{TorsoLhipRhhip}}$, $\theta_{\text{TorsoRhhipLhip}}$, and $\theta_{L\text{hipTorsoRhhip}}$.

In some cases, only using the angle information between structural vectors cannot describe the details of the posture and action. For instance, when determining the relative position between the upper limb and the trunk or head, the angle information cannot provide overall information of the posture. In order to solve this problem, this paper adopts the ratio between three groups of human structural vector modules as auxiliary information to improve the processing of joint point data. An illustration of ratio between human structure vector is shown in Figure 5.

In Figure 5, the a represents the vector from torso to head, the b represents the vector from right hand to head, the c represents the vector from left hand to head, the d represents the vector from torso to left hand, and e represents the vector from torso to right hand. Then, some ratios between human structure vectors are listed in the following equations.

$$\begin{aligned} rat_{ba} &= \frac{\|b\|}{\|a\|} \\ rat_{ca} &= \frac{\|c\|}{\|a\|} \\ rat_{da} &= \frac{\|d\|}{\|a\|} \\ rat_{ea} &= \frac{\|e\|}{\|a\|} \end{aligned} \quad (6)$$

The human posture p refers to the limb state at a specified time. In this paper, the human posture is represented as the combination of processed vector angle and modulus ratio, $p = (p_1, \dots, p_{15})$. The $p_i, i = 1, \dots, 11$ is an angle value, while $p = 12, \dots, 15$ is the modulus ratio. The vector p is the human description vector to depict human posture.

In the research of human behavior recognition, the behavior is usually represented by extracting the posture sequence in the key frame. A continuous posture sequence over a period of time is selected

to represent human action. For action A , $A = (A_1, \dots, A_n)$ in which A_1 is a human posture description vector.

The dynamic time warping (DTW) algorithm is a nonlinear dynamic warping technology that combines distance measurement with time warping method. By using DTW algorithm, it can easily implement the matching between templates with different lengths.

Let $M = \{M(1), \dots, M(m)\}$ represent reference template set, $T = \{T(1), \dots, T(n)\}$ represent text template set. The elements in M and T have the same length. The $d(M(i), T(j))$ represents the bias vector between vector $M(i)$ and vector $T(j)$ to calculate the distortion between $M(i)$ and $T(j)$. The $D(M(i), T(j))$ represents the cumulative distortion which calculates from $i = 1, j = 1$. Then, the $D(M(i), T(j))$ is written as following equation.

$$D(M(i), T(j)) = \sum_{p=1}^{p=j} d(M(p), T(j)) \quad (7)$$

From the Equation (7), it can be found that $D(M(i), T(j))$ is equivalent to calculating the cumulative distortion of a path connecting several eigenvector node pairs.

The purpose of DTW algorithm is to search an optimal path to ensure that $D(M(m), T(n))$ is minimized. The iterative process is implemented through the following equation.

$$D(M(i), T(j)) = d(M(i), T(j)) + D(M(i-1), T(j-1)) \quad (8)$$

In the Equation (8), $D(M(i-1), T(j-1))$ is obtained by the following equation.

$$D(M(i), T(j)) = \min \{D(M(i), T(j-1)), D(M(i-1), T(j-1)), D(M(i-2), T(j-1))\} \quad (9)$$

According to the characteristics of posture description vector, the following similarity calculation formula is adopted in this paper.

$$d(p, q) = \sum_{i=1}^n \frac{(p_i - q_i)^2}{p_i + q_i} \quad (10)$$

In the Equation (10), p and q are two eigenvectors without negative values. In this paper, $p, q \in \mathbb{R}^{15}$. The lower the $d(p, q)$ is, the higher the similarity is. When $d(p, q) = 0$, p complete matches q .

According to the above iterative process, the minimum cumulative distortion between the reference template and the test template can be obtained by repeating recursion from $M(1)$ and $T(1)$ to $M(n)$ and $T(m)$. If the test template is matched with all reference templates one by one, the test template belongs to the category with the smallest cumulative distortion

3. EXPERIMENTS AND SIMULATIONS

In this section, we first collect joint point data, vector angle and modulus ratio of skier in the process of skiing through Kinect camera to collect the regularization processed joint point data. In general,

even for the same behavior of the same person, the distribution of the change trajectory of human joint points in space does not have good aggregation. In addition, massive rotation and translation operations are carried out on the coordinate system, and the data distribution of related nodes does not have good aggregation. The variation trends of vector angle and modulus ratio are roughly the same, and their variation trajectories have good aggregation. The vector angle and modulus ratio obtained after regularization can better meet the invariance of translation and scaling. It provides a good basis for the design of classifier in the next step to recognize the postures of skiers.

The experimental environment of human posture and action recognition is set as follows: the human body faces Kinect camera to ensure that all joint points can be detected, and large-scale movement should be covered by the Kinect camera. We collect six kinds of skiing posture or action, including standing on the ground with skis, standing position on snowboard slope, straight skiing, plow brake, plough turn, and semi plow swing from 50 volunteers. Each volunteer repeats all actions 20 times. Thus, there are 6,000 actions. In each action, 100 sequences are used to establish matching pattern library, while other sequences are used as test set. The corresponding confusion matrix is reported in Table 2.

Table 2. The confusion matrix of the skiing posture and action

	standing on the ground with skis	standing position on snowboard slope	straight skiing	plow brake	plough turn	semi plow swing	recognition rate
standing on the ground with skis	869	26	1	0	0	0	96.99%
standing position on snowboard slope	29	872	8	1	1	1	95.61%
straight skiing	1	0	878	1	0	2	99.55%
plow brake	1	0	10	881	17	9	95.97%
plough turn	0	1	1	13	879	3	97.99%
semi plow swing	0	1	2	4	3	885	98.88%
recognition rate	96.56%	96.89%	97.56%	97.89%	97.67%	98.33%	97.48%

When using DTW algorithm for template matching, the selection of reference template plays an important role in posture or action matching. From the result in Table 2, it can be found that recognition rate of the proposed method can achieve 96.56%, 96.89%, 97.56%, 97.89%, 97.67% and 98.33% for ground with skis, standing position on snowboard slope, straight skiing, plow brake, plough turn, and semi plow swing. The overall recognition rate achieve 97.48% for overall actions.

4. CONCLUSION

In this paper, we propose a framework to recognize the posture and action during skiing sports with the help of Kinect camera. First, the Kinect camera is used to collect joint points of skier's skeleton.

Second, the joint points of skier's skeleton data is represented as the human structure vector which consists of the human intermediate junction angles and modulus ratios. Compared with original joint point data, the intermediate junction angles and modulus ratios have better aggregation for the same behavior. The extracted features are used to construct a matching pattern library. The matching pattern library is used in dynamic time warping (DTW) algorithm to match the future gesture and action. The experimental results show that the proposed framework can identify more 97% skiing postures and actions. The framework can be further used to correct the skiing action and improve athlete's ability.

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Mobile Edge Computing to Assist the Online Ideological and Political Education

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ABSTRACT

With the rapid development of mobile internet technology, mobile network data traffic presents an explosive growth trend. Especially, the proportion of mobile video business has become a large proportion in mobile internet business. Mobile video business is considered as a typical business in the 5G network, such as in online education. The growth of video traffic poses a great challenge to mobile network. In order to provide users with better quality of experience (QoE), it requires mobile network to provide higher data transmission rate and lower network delay. This paper adopts a combined optimization to minimize total cost and maximize QoE simultaneously. The optimization problem is solved by ant colony algorithm. The effectiveness is verified on experiment.

KEYWORDS

Mobile Edge Computing, Online Education, Video Caching Strategy

1. INTRODUCTION

With the development of mobile Internet and Internet of things (IoT) (Srinivasan et al. 2019), mobile HD video (Usman et al. 2018), augmented reality and virtual reality (AR/VR) (Nayyar et al. 2018), and various intelligent hardware devices (Liu et al. 2019; Zhang 2020) have become an indispensable part of people's life. These network technologies and applications do not only enrich people's lives, but also generate huge mobile network traffic.

The rapid growth of mobile network traffic (Yan 2019), especially mobile video traffic, has brought great pressure and challenges to the mobile network. The traffic explosion has brought the following impacts on the current mobile network. First, the pressure of backhaul network and mobile core network is huge. The rapid growth of mobile network traffic makes the pressure of mobile backhaul network increase, the bandwidth resource be tight, and the load of mobile core network is serious. Second, the repeated transmission of content induces in a great waste of network resources. At present, the end-to-end transmission mechanism (Sun et al. 2017) in mobile network will cause the repeated transmission of a large number of popular content, especially the transmission of mobile high-definition video content. Third, the network delay is large which induces bad user experience. In the current mobile network, the user's content request must pass through the base station, S-GW, P-GW to enter the Internet (Zhou et al. 2020). The content is routed to the content server. The spatial distance between the user and the content server (Charu et al. 2017) makes the network transmission delay be larger. In addition, the quality of user experience is influenced on the processing delay of the

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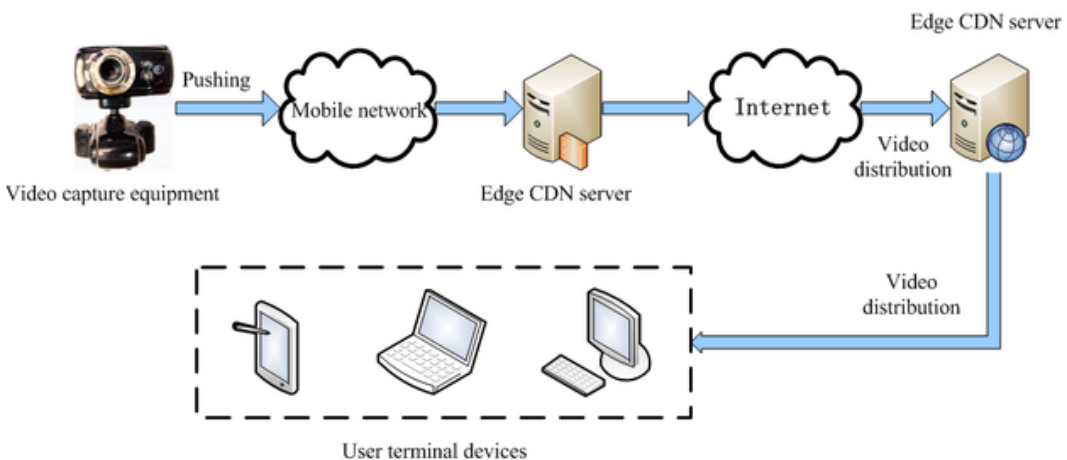
content server, the congestion and packet loss of the transmission link, link failure and other special circumstances (Medeiros et al. 2019).

In order to cope with the explosion of mobile network traffic, improve the quality of users' network experience, accelerate the efficiency of content distribution, and alleviate the transmission pressure of backhaul network, multi access edge computing (Pham et al. 2020) is proposed. It aims to provide end users with ultra-low latency and high bandwidth services through cloud computing capability and service environment for content providers and application developers at the edge of mobile network.

At the same time, the webcast has become an important network application (Na and Jahng 2019). Webcast refers to the process that the audio and video signals are compressed and uploaded to the web server or multimedia server, and distributed on the Internet according to the user's request. In recent years, with the rapid development of the Internet, the webcast has emerged in real life and become popular among mobile users. The current live broadcast services (Wang et al. 2018) include sports events, concerts, remote meetings, online education etc. However, there are still many problems in the current network video live broadcasting system, such as poor mobility, large delay, video jam, difficult to guarantee quality of service.

With the development of MEC technology, video distribution scheme based on MEC has become an important method to improve the quality of webcast. This paper adopts MEC to aid online ideological and political education. The architecture is illustrated in the following figure.

Figure 1. The architecture of MEC based webcast to assist online ideological and political education



2. PROBLEM DESCRIPTION AND PREVIOUS WORK

The content delivery network (CDN) based live video distribution is the mainstream live video distribution scheme (Retal et al. 2017; Viola et al. 2018). The video capture device pushes the collected and recorded video stream to the edge CDN server node which is responsible for caching and transcoding the live video stream. The video stream is distributed to the delivery network server through the Internet or content delivery network to provide live video service for users. In the aspect of mobility, because the edge CDN server is deployed outside the mobile network, when the video capture device is connected to the network through mobile mode, it needs to go through the mobile access network and the core network to access to the CDN server at the edge of the Internet. Thus, it will bring a certain network delay and affect the user experience. When the edge CDN server is

accessed in a fixed way, the mobility of video capture devices will be limited. In addition, the path from the video capture device to the user side edge CDN server is too long and the delay is high. At the same time, the probability of failure on the path is high as well. Thus, the errors are easy to occur. In order to solve this issue, this paper a novel webcast video distribution scheme based on mobile edge computing to assist online ideological and political education.

The live webcast video distribution system based on MEC is composed of video acquisition equipment, base station, MEC server at video source, transmission network, MEC server at user end and user terminal devices. The video capture device first pushes the collected and recorded video to the MEC server. Because MEC has the function of local traffic unloading, MEC can directly push the live video stream to the Internet or distribute the live video stream to the MEC server that is close to the user through the network dedicated line.

Since MEC is generally deployed at the edge of the network close to users in a distributed manner, strengthening the cooperation between distributed MECs is of great significance to improve the efficiency of content distribution. In the cooperative distribution scheme of distributed deployment MEC, there are two connection modes between MECs: through network dedicated line connection and through ordinary Internet connection. The MECs are connected through network dedicated lines, which has higher network bandwidth and can guarantee quality of server (QoS). In the scenario of distributed deployment of MEC, when the video capture device pushes the content to its nearest MEC server, the MEC server immediately synchronizes the content with the MEC server in other locations, and then the terminal device can request the content from the nearest MEC server.

In order to more intuitively describe the proposed webcast video distribution scheme in this paper, we describe the specific distribution process as follows. After the video acquisition device collects the video, it first transmits the video content to the base station through the wireless link, and then the base station directly pushes the video content to the nearest MEC server which caches and transcodes the video pushed by the base station. If the local user requests video content, MEC server can directly respond to the user's request. For non-local users, we introduce the cooperation mechanism of MEC server, that is, the video content on MEC server is synchronized with MEC servers in other places in real time through the network dedicated line or Internet.

Compared with the traditional video distribution scheme, the mobile network live video distribution scheme based on MEC has the following advantages.

- It can ensure video quality. In the traditional scheme, the quality of video content received through web server is affected by uncertainty factors, such as the distance between the video capture device and the web server or CDN server, link congestion and network node failure in the process of video uploading. In the live broadcast scheme based on MEC, the video capture device directly pushes the video content to the MEC server which is deployed at the edge of the network. Thus, it can alleviate this uncertainties in traditional scheme.
- It can reduce transmission delay. In the traditional scheme, the video content needs to be transmitted from the video capture device to the web server or CDN server, and then distributed to the end users in need. However, the long-distance upload from the video capture device to the service server will lead to a relatively large delay. At the same time, the web server or CDN server is not as close to the users as the MEC server. In the MEC based scheme, the MEC server is located at the edge of the network and close to the live video collection device and user. Therefore, the delay of MEC based scheme is lower than that of traditional scheme.
- It implemented link awareness and video online transcoding. In large-scale network video live broadcasting, it requires HD video, such as 4K / 8K HD video. In the traditional scheme, only the video with the highest resolution is transmitted from the video capture device to the web server. For the web server of the content provider, the video is transcoded to the corresponding resolution according to the needs of the user terminal devices. Thus, the traditional scheme does not only require high performance of the web server of the website, but also occupies a large amount of

core network bandwidth. In the MEC based scheme, the MEC server can sense the user link. When it detects that the link of some users is idle, it can recover the link resources and allocate them to other users to transmit high-quality video for users. When it is detected that the user link condition drops, the MEC server can also use its strong data processing or computing power to transcode HD video into video with lower bit rate in real time to meet the needs of end users

The comparison between traditional scheme and MEC based scheme is listed in the following table from the aspects of node number, deployment location, content distribution mode, mobility and distribution path.

Table 1. The comparison between traditional scheme and MEC based scheme

	CDN based scheme	MEC based scheme
Number of nodes	less	more
Deployment location	Deployed in Internet which is far away terminal user	Deployed at the edge of mobile network which is close to terminal user
Content distribution mode	From web server in Internet to terminal users	From video broadcast source to terminal users
Mobility	Less mobility	can support mobility
Distribution path	Fixed network	At the MEC at the edge of the mobile network, it unloads the traffic to the Internet or distribute it through the network dedicated line

3. CACHING STRATEGY FOR VIDEO DISTRIBUTION IN MOBILE EDGE COMPUTING FRAMEWORK

The content caching and transcoding on MEC nodes use the Internet service providers (ISPs) to reduce the cost through reducing the traffic between the return network and ISPs. The video processing at edge nodes can improve the QoE through reduce latency and network congestion at clients.

Let $V = \{V_1, \dots, V_n\}$ represent video set which can be requested by users. Each video V_i has l version with different bit rate. Then, the video is represented as $V_{i,1}, \dots, V_{i,l}$ in which the items are sorted according to bit rate in ascending order. The $V_{i,1}$ represents the lowest bit rate version, while $V_{i,l}$ represents the highest bit rate version. The $V_{i,j}$ represents the version with the j^{th} bit rate. Let $r_{v_{i,j}}$ represent the j^{th} bit rate of video V_i . Let t_i be the length of video V_i . Then, the size of video $v_{i,j}$ is represented as $s_{v_{i,j}} = t_i * r_{v_{i,j}}$.

The MEC server node has caching and computing capabilities. It cannot only cache different video clips with different bit rate versions, but also realize transcoding between different bit rate versions of the same video clip. In this case, when the request for video segment v_i with bit rate $r_{i,j}$ reaches the MEC server node, there are three possible modes.

- Direct hit mode. When there is a video segment v_i in the MEC server node, the MEC server node will respond directly to the user by delivering the video segment v_i .

- Transcoding hit mode. If the video segment v_i does not exist in the MEC server node and there is a high bit rate version $v_{i,j}$, the transcoding mechanism is triggered. The MEC server node can transcode the high version video segment to the requested bit rate version, or request the original video segment from the source content server through the return link, which depends on the decision of the transcoding scheduling scheme. Generally, the higher bit rate version can be transcoded to the lower bit rate version, but the lower bit rate version cannot be upgraded to the higher bit rate version.
- Miss mode. The MEC server node contains neither the video segment v_i of the requested bit rate version nor the higher bit rate version that is available for transcoding. In the Miss mode, the MEC server node will directly obtain the requested video content from the original server to respond to the user's request.

For each video v_i , we use a binary variable $\{x_{v_{i,l}}, x_{v_{i,h}}, y_{v_{i,l}}, z_{v_{i,l}}, z_{v_{i,h}}\} \in \{0, 1\}$ to denote the decision vector when the video request reaches the MEC server node.

The $x_{v_{i,l}}$ is the cache variable to identify whether the video clip with bit rate $r_{i,l}$ has been cached. When $v_{i,l}$ is cached in the MEC server node, $x_{v_{i,l}} = 1$; otherwise, $x_{v_{i,l}} = 0$. Let m represent the maximum of the cache size. The size of video cannot exceed m . The $x_{v_{i,l}}$ satisfies the following condition.

$$\sum_{i=1}^n \sum_{j=1}^l x_{v_{i,j}} * t * r_{v_{i,j}} \leq m \quad (1)$$

The MEC server node can provide computing power to realize transcoding between different bit rate versions of the same video clip. The $x_{v_{i,h}}$ represents whether there exists a video with high bit rate. When there exists high bit rate video, $x_{v_{i,h}} = 1$; otherwise, $x_{v_{i,h}} = 0$. $x_{v_{i,h}}$ can be obtained by the following formula.

$$x_{v_{i,h}} = \min \left\{ 1, \sum_{j=l+1}^l x_{v_{i,j}} \right\} \quad (2)$$

The $y_{v_{i,l}}$ is used to describe the transcoding decision for video $v_{i,l}$. When $y_{v_{i,l}} = 1$, the MEC server node obtains the video through transcoding; otherwise, the video is obtained from source server directly. Because the transcoding can only be performed when the high bit rate version is cached in the MEC server node, the $y_{v_{i,l}} = 1$ is subject to the following constraint.

$$y_{v_{i,l}} \leq x_{v_{i,l}} \quad (3)$$

The $z_{v_{i,l}}$ indicates whether the video is obtained from source server, while $z_{v_{i,h}}$ indicates whether the video is obtained from the adjacent MEC server node. It is assumed that the video is obtained from adjacent server with 50%. Then, $z_{v_{i,h}}$ is constrained by the following condition.

$$z_{v_{i,h}} \leq \frac{rand(\quad)}{0.5} \quad (2)$$

Since $x_{v_{i,l}}$, $x_{v_{i,h}}$, $y_{v_{i,l}}$, $z_{v_{i,l}}$, and $z_{v_{i,h}}$ are binary variables, they must meet the following constraints.

$$\begin{aligned} x_{v_{i,l}} &\in \{0,1\} \\ y_{v_{i,l}} &\in \{0,1\} \\ z_{v_{i,l}} &\in \{0,1\} \\ z_{v_{i,h}} &\in \{0,1\} \\ x_{v_{i,l}} + y_{v_{i,l}} + z_{v_{i,l}} + z_{v_{i,h}} &= 1 \end{aligned} \quad (5)$$

In the transcoding hit mode, there are two options to meet the user's request. Considering the constraints of the cache capacity and computing power of the MEC server, we consider content placement and transcoding scheduling to improve the overall performance of the system. If the system transcodes $r_{v_{i,h}}$ as $r_{v_{i,l}}$, the difference between $v_{i,h}$ and $v_{i,l}$ is represented as $s_{v_{i,h}} - s_{v_{i,l}} = (r_{v_{i,h}} - r_{v_{i,l}}) * t$. The transcoding cost is determined by three factors: the input bit rate, the target bit rate and the video length. The transcoding cost is proportional with the size of the processed data. It needs to balance the transcoding cost and loopback cost.

On the other hand, it can be seen that the user's QoE experience is strongly related to the bit rate version and pause time of the video. The pause time is determined by the response time of the video request which includes the transmission delay and the processing delay in the MEC server. Since we consider the trade-off between various strategies, the processing delay mainly considers the impact of transcoding delay. Directly caching the videos to increase hit rate can obtain fast response time. We prefer to cache more high bit rate video content in the MEC server. However, due to the limited storage size of the edge server and the large video segment of the high bit rate version than that of the low bit rate version, it will occupy a larger cache to result in the reduction of the number of cached videos and increase the overall transcoding delay and transmission delay. In fact, affected by the different network environment of users, the video segments of low bit rate version are more likely to be requested by more users. We should weigh and compromise the caching and transcoding of video segments with high bit rate and high popularity based on the popularity of different bit rates.

In order to effectively enhance the overall performance of the system and the overall QoE of users, we need to comprehensively consider the system cost and QoE, and schedule video requests through caching strategy and transcoding strategy. We design the joint optimization problem of cache and transcoding of MEC sever through user request model, delay model, QoE model and cost model.

The probability distribution vector of the video segment set is represented as $\{p_1, \dots, p_n\}$ ($\sum_{i=1}^n p_i = 1$). The p_i represents the probability of that video v_i is requested. If the probability follows Zipf distribution, the probability can be obtained by the following equation.

$$p_i = \frac{k}{i^\alpha}, \quad k = \frac{1}{\sum_{i=1}^n \frac{1}{i^\alpha}}, \quad v_i \in V \quad (6)$$

In equation (6), α represents skewness level in the popularity curve ($\alpha > 0$). The same video with different bit rate follows normal distribution. Then, the probability of l^{th} version of video is rewritten as follows:

$$p_i = \frac{e^{-\frac{(l-m)^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} \quad (7)$$

In Equation (7), m and σ^2 are mean and variance, respectively. The probability of that video v_i with l^{th} bit rate is written as follows:

$$p_{v_i,l} = p_{v_i} * p_l \quad (8)$$

The quality of experience (QoE) plays an important role in the Internet video applications. The QoE is effected by the experience quality perceived by the adaptive video streaming client, startup delay, average playback bit rate, and bit rate switching. The video quality depends on the video bit rate. The video bit rate is directly related to the perceived quality of video. Higher compression produces a smaller bit rate but lower perceptual quality. It still needs to balance video bit rate and pause rate. The high bit rate streams increase the possibility of experiencing pause events, since there is a higher chance that the download throughput will be reduced below this bit rate due to contention on the bottleneck link or reduced wireless link quality. Low quality streaming reduces the possibility of pause, but also significantly reduces the quality of the customer experience.

The startup delay is the duration of time that the client needs to reach the target buffer fill level. It is associated with the waiting time from the click of the client to the start of playback. The impact of startup delay on audience dissatisfaction is significantly less than that of stall event.

Stall ratio refers to the total duration time of the session to stop video playback. Low download throughput will lead to a pause event when the playback buffer is empty. It is important to avoid stall events because they play an important role to effect QoE. When the total amount of resources is enough to support the minimum available video bit rate of all clients, it needs to avoid pause events as much as possible.

Frequent quality transcoding is harmful to QoE. It seems that q_{v_i} represents the QoE to associate with video v_i . The QoE of whole system q_{sys} can be represented by the weighted sum of the corresponding measures of all video blocks QoE..

$$q_{sys} = vl \sum_{i=1}^n \sum_{j=1}^l p_{v_i,j} q_{v_i,j} \quad (9)$$

The video quality q_{v_i} is determined by the bit rate r_{v_i} . However, the bit rate does not directly reflect the video quality. It needs to map the bit rate to the video quality through the following equation.

$$q_{v_i} = \ln(r_{v_i} + 2) \quad (10)$$

The delay constraint is one of the main challenges that video service providers should face. The audience suffers network delay, transcoding delay and broadcast delay. In this paper, we focus on the delay caused by the return link from the source server or adjacent MEC server node to the destination MEC server node. The associated cost includes caching, transcoding, and transmission. Let w_c represent the caching cost at MEC server node per video. The caching cost for video v_i is represented as follows:

$$c_v = w_c * t * r_v \quad (11)$$

The transcoding cost is represented as follows:

$$c_s = w_c * D_s = w_s * cm * t * \frac{r_{v_h} - r_{v_l}}{f_0} \quad (12)$$

In Equation (12), D_s represents transcoding delay. Let w_{t0} and w_{t1} represent the transmission cost of transmission link from MEC server provider to source server and the link from MEC server provider to adjacent MEC server node. The associated cost is rewritten as follows:

$$c_{t0} = w_{t0} * t * r_v \quad (13)$$

$$c_{t1} = w_{t1} * t * r_v \quad (14)$$

The total cost is represented as following equation.

$$c_{total} = c_v + c_s + c_{t0} + c_{t1} \quad (15)$$

In order to implement cache transcoding resource scheduling with low cost and high QoE, we define the optimization problem as follows:

$$\begin{aligned} \min_{x_{v_l}, y_{v_l}, z_{v_l}, z_{v_h}} : & \alpha * c_{total} - \beta * q_{total} \\ & \sum_{i=1}^n \sum_{j=1}^l x_{v_{i,j}} * t * r_{v_{i,j}} \leq M \\ & \sum_{i=1}^n \sum_{j=1}^l y_{v_{i,j}} * cm * t * (r_{v_{i,h}} - r_{v_{i,l}}) \leq f_0 \\ & y_{v_{i,j}} \leq x_{v_{i,j}} \\ & x_{v_l} \in \{0,1\}, \quad y_{v_l} \in \{0,1\}, \quad z_{v_l} \in \{0,1\}, \quad z_{v_h} \in \{0,1\} \\ & x_{v_l} + y_{v_l} + z_{v_l} + z_{v_h} = 1 \end{aligned} \quad (16)$$

The objective function in optimization problem (16) is to minimize the total cost as well as maximize total QoE. It covers caching, transcoding, and transmission delay and caching, transcoding, and transmission cost.

The combination of caching and transcoding is a integer linear programming which is a NP-hard problem. It can be solve by an evolution algorithm to obtain an approximate solution, such genetic algorithm (Whitley 1994), ant colony algorithm (Dorigo 2006), and simulated annealing algorithm (Rutenbar 1989). In this paper, the ant colony algorithm is adopted to solve Equation (16).

4. EXPERIMENTS AND SIMULATIONS

In this section, we establish a simulation environment to verify the proposed caching and cost optimization strategy for mobile edge computing architecture in online education. For the video cache transcoding process, we focus on the following three strategies. The first strategy considers caching, transcoding and backhaul. When a transcoding hit occurs, the mobile edge computing server can respond to the user by transcoding or requesting video content from the source server or the adjacent mobile edge computing server.

The second strategy only considers backhaul in which the video requests can only be transmitted through the network to the source server for response.

The third strategy considers caching and transmission in which the mobile edge computer server only has caching function and does not have transcoding capability.

The results of total QoE with the number of videos are reported in Figure 2, while the results of total cost with the number of videos are reported in Figure 3.

Figure 2. The curve between the number of videos and the total QoE

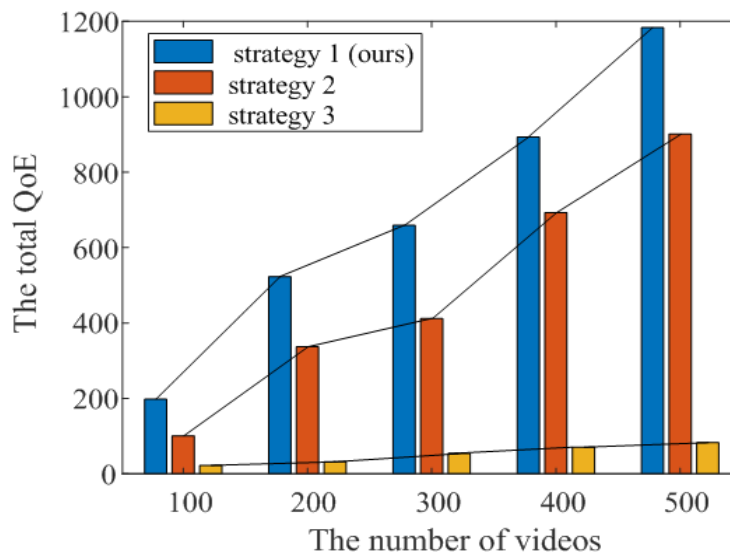
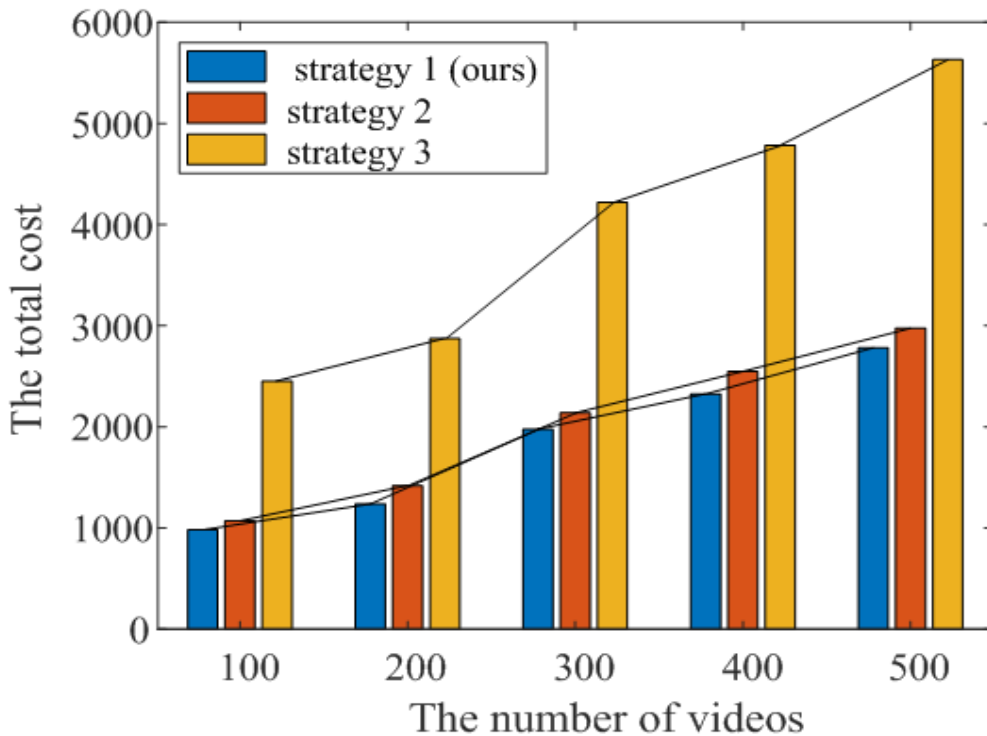


Figure 3. The curve between the number of videos and total cost



From results in Figure 2, it can be found that the total QoE increases with the number of videos in three strategies and the proposed strategy can obtain better total QoE than both strategy 2 and strategy 3. From the results in Figure 3, it can be found that the total costs increases with the number of videos in three strategies and the proposed strategy cost fewer than both strategy 2 and strategy 3.

5. CONCLUSION

In this paper, we establish a mobile edge computing (MEC) based live video distribution system to support online education. Compared with content delivery network (CDN) based live video distribution, the MEC based live video distribution system has better mobility. The cache transcoding resources in MEC based live video distribution is optimized by the joint optimization which comprehensively considers the overall cost consumption on the network side and the overall QoE optimization on the user side. The joint optimization can effectively schedules the cache, transcoding and return strategies of video content to improve the overall performance of the system. The proposed optimization is solved by ant colony algorithm. The experimental results show that the proposed strategy can significantly improve the performance of the live video distribution system.

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Online Music Style Recognition via Mobile Computing

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ABSTRACT

Music is a widely used data format in the explosion of internet information. Automatically identifying the style of online music in the internet is an important and hot topic in the field of music information retrieval and music production. Recently, automatic music style recognition has been used in many real-life scenes. Due to the emerging of machine learning, it provides a good foundation for automatic music style recognition. This paper adopts machine learning technology to establish an automatic music style recognition system. First, the online music is processed by waveform analysis to remove the noises. Second, the denoised music signals are represented as sample entropy features by using empirical model decomposition. Lastly, the extracted features are used to learn a relative margin support vector machine model to predict future music style. The experimental results demonstrate the effectiveness of the proposed framework.

KEYWORDS

Empirical Model Decomposition, Mobile Computing, Music Style Recognition, Relative Margin Support Vector Machine, Waveform Analysis

1. INTRODUCTION

With the rapid development of the Internet, people begin to have the opportunity to come into contacting with different music styles around the world and enjoy the pleasure brought by music (Kohut 2018). Although the music has some different expression ways in different countries or nations, the music has always been able to express people's thoughts, convey people's thinking (Stewart 2019). The music fully expresses its value in human life.

Up to now, there are countless kinds and quantities of musical instruments, such as guitar, piano, erhu, and cello (Zhao 2019). At the same time, the storage methods of music files are becoming diversification, including WAV files, MIDI files, MP3 files and WMA files (Fu 2021). Each file format has its own storage characteristics. WAV files are easy to generate and edit, but the compression ratio is not high. MIDI file has small size and storage space and is especially adapted for the needs of long time music. MP3 file is also a very common music format, which has high compression rate and poor sound quality.

The emergence of musical instruments and the diversification of music storage methods promote the formation of music genres (Ojanen 2020). The common music genres include jazz, classical, pop, hip-hop, rock and roll, etc. Recently, music information retrieval has become a new research topic, and a large number of researchers have studied this field (Murthy 2018). Because some users are only interested in one genre of music, music recognition and classification system can divide music

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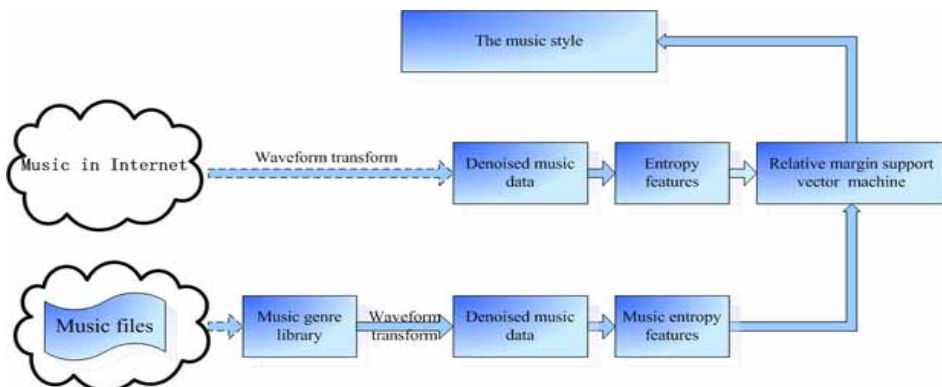
into different genres. In this way, it is convenient for users to search and manage music in different periods of time, such as sports and rest. When the same song is sung by different people, the songs are different due to the different range, timbre and the performance of various musical instruments. Exactly extracting features of music leads to the low efficiency of music genre classification and recognition. With the deepening of a large number of research works, the music genre classification and generation will gradually develop in a better direction.

With the rapid development of music storage and computer technology, digital audio processing technology has been greatly improved (Gao 2019). Music processing software is also diversified, such as windows recorder, tools when buying a sound disc (Liang 2021). Speech recognition, text to speech conversion, speech compression coding and other technologies have gradually begun to use computer technology instead of manual completion (Lero 2019; Delić 2019). These methods are based on digital audio processing. In general, a piece of music is vectored. The features of the music are represented as a series of numbers to store and transmit. Different formats of music are finally represented in digital form, which makes the analysis of music become very simple and efficient by combining with the music processing software in the computer. At the same time, with the rapid development of computer technology, the accuracy of analyzing and processing music has been greatly improved, and the cost is reduced.

In the field of music style recognition, the most important thing is the extraction of related features and the selection of classifiers (Ghosal 2018). When selecting different music feature vectors for music style recognition, different classification results will be produced. At present, the commonly used features are tone, timbre, loudness etc. (Prince 2019) In order to improve the effect of style recognition, researchers have taken many methods, such as signal processing method (Wang 2019). These methods improve the effect of music classification to a certain extent. However, the feature extraction is still difficult under some special cases. In order to further improve the performance of music style recognition, it is necessary to deeply mine the internal association between music signals.

With the development of machine learning and deep learning in face recognition, speech recognition and image recognition, people gradually try to apply machine learning and deep learning to the field of music generation and recognition. The deep learning is widely used in storing and processing large amounts of data, such as recurrent neural network (Purwins 2019). However, we cannot get the data of the previous moment or earlier, which makes the effect of classification or the acquisition of music features, such as tone, timbre, loudness, rhythm, be inaccurate. This paper adopts a robust support vector machine to avoid the issue that training set needs massive samples in deep learning for online music style recognition. The flowchart of music style recognition is summarized as following figure.

Figure 1. The architecture of music style recognition



2. PROBLEM DESCRIPTION AND PREVIOUS WORK

The music style recognition refers to classify electronic music according to the associated features. In essence, it is a process of pattern classification, which is a very complex process and involves many disciplines, such as psychology, signal processing, pattern recognition, etc. The format of electronic music in the network is different from that in real life. The electronic music has its own special format. The scholars put forward a content-based electronic music recognition model, which extracts the mean value and autocorrelation coefficient of electronic music as features, and then establishes an electronic music recognition model (Hong 2018).

Different genres have different music styles. Music style or music genre recognition has been widely studied for a long time (Vishnupriya 2018). Since the 1990s, people began to use artificial methods to determine the genre and style of music. The famous “music chromosome project” asks music experts divide music into different types through their understanding and technology of music. However, due to the limited technical conditions, the artificial method is very immature when facing massive music data. Additionally, different experts have slightly bias on understanding of different music genres. This project spends a lot of financial and material resources, but the result is not satisfactory. Driven by this situation, people began to study the automatic classification algorithm of music genre style recognition.

In 1996, American researchers proposed a classification algorithm, which calculates the mean, variance and autocorrelation coefficient from massive music data. These features are used to further analyze the characteristics of music, such as loudness, tone and other features that people can easily feel. Then, music styles are determined by classifiers. Many improvements are based on these methods. For instance, the acoustic features, including timbre, rhythm, and pitch, were introduced to represent the features of music. However, due to high dimensionality of these features, the feature selection algorithm is used to reduce the dimension of features, which can reduce the calculation and remove the irrelevant redundant information.

With the development of computer science and artificial intelligence technologies, machine learning has been applied to music genre recognition, such as k-nearest neighbors, conditional random field, linear discriminant analysis, Gaussian mixture model and Markov model. With the application of wavelet transform theory, the statistical method is introduced to calculate the statistical value of wavelet coefficients which are used to combine with the classification models in machine learning, such as linear discriminant analysis, Gaussian mixture model, Markov model. In order to further improve the essential features of electronic music, the unsupervised dimensionality reduction is introduced into music style recognition which further improves the music representation. Another way is to introduce high order matrix features to represent electronic music which can improve the effect of classification and recognition. However, the dimensions increase significantly, which makes the calculation process very complicated and the training be very time-consuming.

With the development of deep learning, more and more researches pay attentions to applied deep learning in music generation, such as long short term memory network, recurrent neural network. Generally, the long short term memory network is superior to recurrent neural network for music style recognition. The reason is the difference of the structure. Compared with deep learning, Markov model is relative simple. However, when applying Markov model to music generation, it cannot get the temporary structure for a long time. The Markov model can only learn some simple music examples and quality of produced music is not very good. Generative adversarial nets (GAN) is not very good in dealing with the variable content. One of the problems is that the generated music will add some rhythms different from the original music in it. The limited Boltzmann machine is slightly less controllable.

3. FEATURE EXTRACTION AND MUSIC STYLE RECOGNITION

The wavelet analysis can decompose the one-dimensional signal with different resolutions and refine the original signal (Balasubramanian 2018). In this paper, the wavelet analysis is used to remove the noise and improve the signal-to-noise ratio to facilitate the subsequent signal processing. The signal decomposition form can be described as following equation.

$$\begin{cases} x_k^j = \sum h_0(n-2k)x_n^{j+1}, & j \geq 0 \\ d_k^j = \sum h_1(n-2k)x_n^{j+1}, & j \geq 0 \end{cases} \quad (1)$$

In Equation (1), x_k^j represents the low frequency part of the original signal, d_k^j represents the high frequency part of the original signal, h_0 is a low pass filter, h_1 is a high pass filter, N is the wavelength of the filter, k is the scale of wavelet decomposition. Then,

$$k_1(n) = (-1)^{-n} h_0(N-n) \quad (2)$$

When the original signal after multi-layer decomposition, it can get different levels of coefficients. By using a denoising algorithm to process all levels of wavelet coefficients (such as soft threshold algorithm), the new signal can be represented as follows:

$$\hat{x} = T_h(\gamma, t) = \begin{cases} \text{sgn}(\gamma)(|\gamma| - t), & |\gamma| \geq t \\ 0, & |\gamma| < t \end{cases} \quad (3)$$

In Equation (3), $\hat{x}$ represents the waveform coefficients, t represents the threshold, $t = \sigma\sqrt{2\log N}$.

The signal processed by soft threshold algorithm is reconstructed by wavelet analysis as follows:

$$x_n^{j+1} = \sum_k h_0(n-2k)x_k^j + \sum_k h_1(n-2k)d_k^j \quad (4)$$

The original signal is decomposed and reconstructed by wavelet analysis. The noise in the original signal can be effectively eliminated, which can greatly improve the signal-to-noise ratio of the signal. The denoised signal is further processed by empirical model decomposition (Wei 2018).

Empirical mode decomposition can decompose a signal into several intrinsic model function (IMF) components and residual components

$$x(t) = \sum_{i=1}^c IMF_i(t) + r(t) \quad (5)$$

In Equation (5), c represents the number of intrinsic mode function components, IMF_i represents the i^{th} intrinsic mode function component, $r(t)$ represents residual component.

Let $\{x_1, \dots, x_n\}$ represent a music signal sequence, m represent the number of dimensions of music signal, r represent similarity tolerance. The sample entropy can be computed by the following procedure.

Step (i). The music signal sequence is used to construct a vector according to the sequence number as follows:

$$X(i) = \{x(i), x(i+1), \dots, x(i+m-1)\} \quad (6)$$

Step (ii). The distance between two sequences $X(i)$ and $X(j)$ as $d(X(i), X(j))$ as follows

$$d(X(i), X(j)) = \max_{k=0 \rightarrow m-1} \{X(i+k) - X(j+k)\} \quad (7)$$

Step (iii). Counting the number of signals with $d(X(i), Y(j)) < r$ for $i = 1, \dots, N-m+1$, $j = 1, \dots, N-m+1$. The N represents the length of signal. Then, the ratio between n_i^m and the sum of distance $N-m$ is represented as follows:

$$C_i^m(r) = \frac{n_i^m}{N-m} \quad (8)$$

Step (iv). Let $m = m+1$, repeating Step (i) to (iv) to compute $C^{m+1}(r)$.

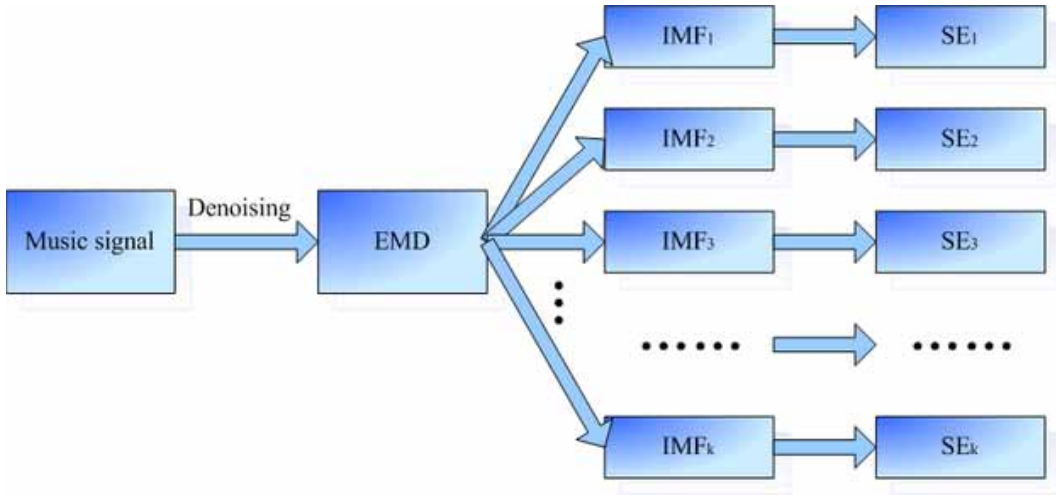
Step (v). The sample entropy is represented as follows:

$$SE = -\ln \frac{C^{m+1}(r)}{C^m(r)} \quad (9)$$

The parameters m and r are close to the sample entropy. In general, they can set according to user experience.

The denoised music signals are decomposed by empirical model decomposition to obtain intrinsic mode function components. Then, the sample entropy is computed according to above procedure. The music style features are represented as $[SE_1, \dots, SE_k]$. Here, SE_k represents the sample entropy of the k^{th} intrinsic mode function component. The whole feature extraction of music style is summarized as following figure.

Figure 2. The architecture of feature extraction for music signal



The electronic music is represented as sample entropy features. Then, the sample entropy features of all electronic music are used to construct a training set. Then, the training set is used to learn a classifier to predict future music style.

The support vector machine is a widely used classification algorithm. However, it treats every samples in training set equally and is not robust to the noises in the training set. In order to solve this issue, instance-weighted support vector machine (Zhu 2016) and relative margin support vector machine (Zhu 2017) are proposed. In this paper, we adopt relative margin support vector machine as the classification algorithm in music style recognition.

Let $\{X, Y\}$ represent the training set consisting of $\{x_i, y_i\}$ ($i = 1, \dots, l$) in which x_i represents music sample entropy features and y_i represents the music style. When there are only two styles, $y_i \in \{+1, -1\}$. The aim of relative margin support vector machine is to find a hyperplane $f(x) = w^T \phi(x) + b$ via maximizing the minimum relative margin between two classes. The $\phi(x_i)$ represents the mapping of sample x_i in reproducing kernel Hilbert space (RKHS). The mapping function $\phi(x)$ is an implicit function. However, the inner product between mappings $\phi(x_i)$ and $\phi(x_j)$ can be easily computed by a kernel function, $\langle \phi(x_i), \phi(x_j) \rangle = K(x_i, x_j)$. The common used kernel function includes polynomial kernel, Gaussian kernel, hist kernel etc. When the linear kernel is used, the hyperplane is degraded as $f(x) = w^T x + b$.

The hyperplane of relative margin support vector machine can be obtained by solving the following optimal programming.

$$\begin{aligned}
 \min_{\mathbf{w}, \rho} \quad & \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^l \xi_i \\
 s.t. \quad & y_i \left(\mathbf{w}^T \phi(\mathbf{x}_i) + b \right) \geq \eta_i - \xi_i, \quad i = 1, 2, \dots, l \\
 & \xi_i \geq 0, \quad i = 1, 2, \dots, l.
 \end{aligned} \tag{10}$$

Due to Equation (10) is a convexity problem, the solution can be obtained through the associated dual programming. The η_i represents the relative margin of sample x_i , which is determined by prior

knowledge of the training set. The η_i is defined by relative density degree. The paper adopts prior knowledge of the training set by using nearest neighbors' distribution.

By introducing the Lagrange multipliers α_i for constraints $y_i(\mathbf{w}^T \varphi(\mathbf{x}_i) + b) \geq \eta_i - \xi_i$ ($i = 1, \dots, l$) and β_i for constraints $\xi_i \geq 0$ ($i = 1, \dots, l$), the Lagrange function is written as follows:

$$L = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^l \xi_i - \sum_{i=1}^l \alpha_i \left(y_i (\mathbf{w}^T \varphi(\mathbf{x}_i) + b) - \eta_i + \xi_i \right) - \sum_{i=1}^l \beta_i \xi_i \quad (11)$$

Let the partial derivatives of L for w , ξ_i , b be equal to 0, then the following equations hold

$$\mathbf{w} = \sum_{i=1}^l \alpha_i y_i \varphi(\mathbf{x}_i) \quad (12)$$

$$C - \alpha_i y_i - \beta_i = 0 \quad (13)$$

$$\sum_{i=1}^l \alpha_i y_i = 0 \quad (14)$$

By substituting Equation (12), (13), and (14) into Equation (11), we can obtain the dual form of Equation (10) as follows:

$$\begin{aligned} \min_{\pm} \quad & \frac{1}{2} \sum_{i,j=1}^l \alpha_i \alpha_j y_i y_j K(x_i, x_j) - \sum_{i=1}^l \alpha_i \eta_i \\ \text{s.t.} \quad & \sum_{i=1}^l y_i \alpha_i = 0 \\ & 0 \leq \alpha_i \leq C, \quad i = 1, \dots, l \end{aligned} \quad (15)$$

The Equation is a quadratic programming which can be solved by many existing algorithms. After obtaining α , the parameters w and b in hyperplane $f(x)$ can be obtained by the following equations.

$$w = \sum_{x_i \in SVs} \alpha_i y_i \varphi(x_i) \quad (16)$$

$$b = y_i \eta_i - \sum_{j=1}^l \alpha_j y_j K(x_i, x_j), \quad 0 < \alpha_i < C \quad (17)$$

In Equation (16), the SVs represents the support vector set which consists of the samples with nonzero Lagrange multipliers. The final hyperplane is written as follows:

$$f(x) = \text{sgn} \left(\sum_{x_j \in SVs} \alpha_j y_j K(x_i, x_j) + b \right) \quad (18)$$

The relative margin support vector machine can solve the two-class classification problem. For multi-class classification problem, it still needs to adopts one-to-one strategy or one-to-rest strategy to convert the multi-class classification problem as several two-class classification problems. For multi-class classification problem, the whole procedure of relative margin support vector machine is summarized as following algorithm.

Algorithm (relative margin support vector machine for multi-class classification)

- Step 1:** converting the multi-class classification problem via one-to-one strategy or one-to-rest strategy as $c(c-1)/2$ two-class classification problems for one-to-one strategy or c for one-to-rest strategy;
- Step 2:** calculating relative margin parameter η_i ;
- Step 3:** solving relative margin support vector machine by Equation (15) for each two-class classification problem;
- Step 4:** calculating values of Equation (16) for all samples in negative class.

In Step 3 of reduced support vector machine, l_p represents the number of samples in positive class. In Step 6 of reduced support vector machine, l_n represents the number of samples in negative class. The parameter δ represents the size of retained subset of the training set. The larger δ is, the more samples are retained in reduced support vector machine.

4. EXPERIMENTS AND SIMULATIONS

This section first adopts GTZAN dataset (Sturm 2013) to evaluate proposed music style recognition method. Then, the proposed method is used to recognize the styles of online music. The GTZAN is a western music genre library, which is widely used to evaluate music style recognition methods. The GTZAN music genre library contains 10 music genres, including blues, classical, country, disco, hip hop, jazz, metal, pop, reggae and rock. Each music genre contains 100 audio tracks which last 30 seconds. Each track is with 22,050 Hz Mono 16-bit and stored in the form .wav format. Each track is split as three parts. Each part lasts 10 seconds. Then, there are 3,000 music clips. The clips are divided as two parts as the training set and test set, respectively. One part contains 2,500 clips (250 per music genre) as training set, while the other part contains 500 clips (50 per music genre) as test set.

In order to ensure the relative margin support vector machine can solve the multi-class classification problem, the one versus rest strategy is adopted in the relative margin support vector machine. The radial basis function is used as the kernel function. The recognition result is reported in the form of confusion matrix in the following table.

Table 1. The confusion matrix of music style recognition for GTZAN dataset

	blues	classical	country	disco	hip hop	jazz	metal	pop	reggae	rock	Acc. (%)
blues	41	4	0	0	0	4	1	0	3	0	77
classical	3	38	1	0	0	5	0	0	0	0	81
country	0	3	44	1	0	1	1	0	0	0	88
disco	2	2	2	39	2	0	0	4	4	7	63
hip hop	0	0	0	1	38	0	0	0	1	0	95
jazz	3	3	0	1	0	37	1	0	1	0	80
metal	1	0	3	0	1	0	43	0	1	0	88
pop	0	0	0	7	8	3	0	41	0	5	64
reggae	0	0	0	0	0	0	0	1	40	3	91
rock	0	0	0	1	1	0	4	4	0	35	78
Acc. (%)	82	76	88	78	76	74	86	82	80	70	79

From the results in Table, it can be found that the proposed method can identify 79% of the music genre in the test set. In Table 2, we compare the proposed method with some previous methods, including decision tree, k nearest neighbors, support vector machine, and linear discriminant analysis. In order to eliminate the randomness when spitting training set and test set, the experiments are repeated 20 times and the results in Table 2 are the mean of the twenty times. The experimental results are reported in terms of accuracy, F1-measure and precision. The best results in each column are highlighted by bold font.

Table 2. The performance comparison of music style recognition with different methods

Methods	Accuracy (%)	F1-mesuaure	Precision (%)
Decision tree	75.83	0.7746	78.91
K nearest neighbors	74.73	0.7639	77.65
Linear discriminant analysis	76.69	0.7929	80.59
Support vector machine	77.89	0.8017	81.07
Ours	79.67	0.8179	82.39

From the results in Table 2, it can be found that the proposed music style recognition method archives 79.67%, 0.8179, 82.39% for accuracy, F1-measure, and precision, respectively. The proposed music style recognition method is superior to decision tree, k nearest neighbors, linear discriminant analysis, and support vector machine.

Furthermore, we also used the proposed music style recognition method to identify the style of the music in Internet. First, the music is crawled from Internet. Then, the crawled music is processed by wavelet transform and EMD to convert as extracted features. Lastly, the extracted features are input into the learnt music style recognition model to determine the associated music style. The predicted style is sent to music expert to determine whether the music is correctly identified. The experimental results show that 76.35% music from Internet can be correctly recognized.

5. CONCLUSION

In this paper, we propose a music style recognition framework based on machine learning and mobile computing technologies. First, a music genre library is constructed by using collected music files. Second, the collected music signals are extracted from these files and denoised by using waveform transform. Third, the denoised signals are processed by using empirical model decomposition to extract sample entropy features. Fourth, the sample entropy features are used to learn a classification model. In this paper, we adopt relative margin support vector machine. Since there are more than two music genres, the one versus rest strategy is adopted in relative margin support vector machine. The music in Internet is represented as associated sample entropy features to input into the learnt classification model. The experimental results demonstrate that the proposed music style recognition framework can identify more than 76% music.

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English Article Style Recognition and Matching by Using Web Semantics

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ABSTRACT

With the explosion of internet information, people feel helpless and find it difficult to choose in the face of massive information. However, the traditional method to organize a huge set of original documents is not only time-consuming and laborious, but also not ideal. The automatic text classification can liberate users from the tedious document processing work, recognize and distinguish different document contents more conveniently, make a large number of complicated documents institutionalized and systematized, and greatly improve the utilization rate of information. This paper adopts termed-based model to extract the features in web semantics to represent documents. The extracted web semantics features are used to learn a reduced support vector machine. The experimental results show that the proposed method can correctly identify most of the writing styles.

KEYWORDS

Document Type Recognition, Reduced Support Vector Machine, Vector Space Model, Web Semantics

1. INTRODUCTION

With the continuous popularization and development of information superhighway, information technology has penetrated into every corner of our social living (He 2021, Camero 2019). It has changed people's life and work style with unprecedented speed and ability. We are in an era of information explosion (Kumari 2017). On one side, the Internet contains a vast amount of information which is far beyond people's imagination. On the other hand, people often feel helpless when they face the vast ocean of information. It is called as information overload (Schmitt 2018, Swar 2017). It is a challenging task to help people effectively to manage massive information and quickly select useful information that they are interested in.

The web information is increasing, including online news, e-magazines, online technical reports, online documents, e-mail, BBS, online announcements (Yamamoto 2018). The traditional method to handle daily huge amount of information is time-consuming and laborious. The automatic text classification can directly filter and classify the document information (Kadhim 2019, Nguyen 2018). The user can only receive minor part which they are interested in. Then, users can be liberated from tedious document processing work and can easily understand and distinguish different document

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contents. A large number of complicated documents can be regulated and systematized, and the utilization rate of information can be greatly improved.

One the other hand, while people can easily send and obtain web information, they also face with many harmful or illegal information, such as pornography, violence, and superstition. Unhealthy content such as gambling can be seen everywhere on the Internet. Even criminal text may be exist in BBS, blog, e-mail etc. to carry out reactionary propaganda, fraud, extortion, terrorist threats, drug sales and other illegal and criminal activities. Installing illegal information filtering software cannot effectively prevent the occurrence of illegal web information. Through legislative means, investigating the criminal responsibility of criminals can effectively crack down on this kind of criminal behavior. However, due to the lack of effective evidence, the criminals may be free from the evasion of legal sanction.

Text classification refers to the process of marking a free document with one or more predefined category labels according to its content information (Kowsari 2019, Mirończuk 2018). In order to correctly perform the task of text classification, it must input the useful information of the text into the computer to scientifically abstract the text and establish mathematical model to describe the text. The document expression is a key part of the text classification. The text representation refers to many representation methods and techniques of text retrieval (Wang 2020, Luo 2019). The common used text retrieval methods include: Boolean model (Lashkari 2009), vector space model (Raghavan 1986) and probabilistic model (Feng 2018). These models deal with feature weighting, category learning, and similarity calculation from different perspectives.

By closely combining with machine learning, the vector space model has been successfully used in text classification and becomes a mainstream method in the field of text classification. Vector space model (VSM) was first proposed the field of information retrieval. Then, it has been widely used in the field of text classification. In the vector space model based text classification method, the documents are converted as vector form by using term frequency and inverse document frequency. The vectors are indexed by inverted documents to calculate document similarity. Although vector space model has been solved text representation, it still needs to assume that words in the document are independent with each other to reduce the complexity of the representation. This paper adopts a termed-based model to represent the document and utilizes the extracted features to learn a reduced support vector machine to recognize the document type.

2. PROBLEM DESCRIPTION

Text classification is a supervised learning process. It learns a classification model to represent the relation between text features and text labels by a training set which consists of massive labeled documents. The features text document are input into learnt classification model to determine the document type. The text classification is mathematical mapping which maps the document to the associated type. The mapping can be represented as:

$$f : T \rightarrow L \quad (1)$$

In Equation (1), T represents the document set, while L represents the document type set.

The mapping rule of text classification is based on the data information of samples from each class to summarize the regularity of classification and establish the rules to determine the text related categories. The classification of text is based on document's content other than the data pattern in the document. It means that the concept of which type of text is related to is subjective.

From the above description, the flowchart of text classification contains two stages: training classification model and predicting future document according to learnt classification model. In the training classification model stage, the documents in the training set are represented as a unified form

by using vector space model. The features are processed by using feature extraction method. The processed features are used to learn a classifier. In predicting future document stage, a new document is represented as a feature vector by using vector space model. The feature vector is processed by using feature extraction method and the processed feature vector is input into the learnt classification model to determine its type. An illustration of the text classification is shown in Figure 1.

In Figure 1, the text classification framework consists of four parts: text preprocessing, text representation, classifier training and classification evaluation. The feedback mechanism heuristic adjusts the parameters of feature extraction and classification algorithms to make the text recognition can achieve the best performance.

3. DOCUMENT REPRESENTATION

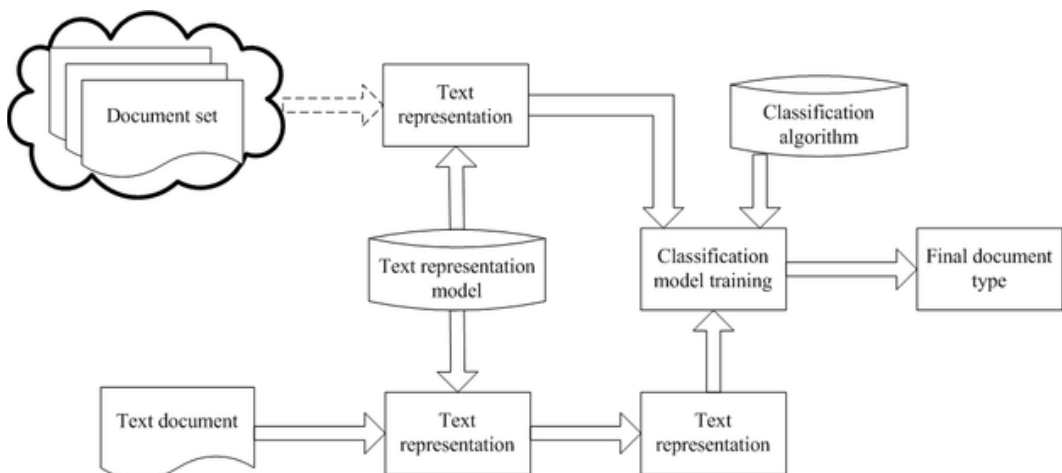
When reading an article or document, we can have a fuzzy understanding of the content according to our own understanding ability and experience. However, the computer does not have this ability. The human understanding refers to text semantics (Sinoara 2017). According to the current research level of computer technology, it is impossible for machine to read the natural text that people can understand. The machine only knows and understands 0 or 1. Thus, it is necessary to convert the document as the form that machine can understand. A unified expression is needed for the document.

With the development of information retrieval technology, several text retrieval models are developed. The current text retrieval models include Boolean model, vector space model, and probability model, etc. These models process feature weighting, category learning, and similarity computation from different perspectives. The vector space model is an effective text representation model.

In vector space model, it only considers the frequency of words and neglects the order of words in the document. The words in document should be different. The document space can be regarded as a vector space composed of a set of orthogonal entry vectors. Each document is represented as one of the normalized eigenvectors which is rewritten as follows:

$$S(d) = \left\{ (t_1, w_1(d)); \dots; (t_i, w_i(d)); \dots; (t_n, w_n(d)) \right\} \quad (2)$$

Figure 1. The architecture of document type recognition framework



In Equation (2), t_i represents word item in document d , $w_i(d)$ represent the weight of word item t_i in document d . The word item t_i can be either the word or the phrase in the document d to improve the accuracy of content feature representation. The $w_i(d)$ can be regarded as a function of the frequency of word item t_i in the document d , which can be represented as $w_i(d) = \phi(tf_i(d))$. The $tf_i(d)$ represents the frequency of word item t_i in document d . The common used function for weight include: Boolean function, square root function, logarithmic function, and TFIDF function, which are written as Equation (3), (4), (5), and (6), respectively:

$$\phi(tf_i(d)) = \begin{cases} 1, & tf_i(d) \geq 1; \\ 0, & tf_i(d) = 0. \end{cases} \quad (3)$$

$$\phi(tf_i(d)) = \sqrt{f_i(d)} \quad (4)$$

$$\phi(tf_i(d)) = \log(tf_i(d) + 1) \quad (5)$$

$$\phi(tf_i(d)) = f_i(d) \times \log \frac{N}{n_i} \quad (6)$$

Here, N represents the number of documents, while n_i represents the number of documents which contain word item t_i .

In general, the words, phrases, concepts are adopted as feature words. The word is the smallest semantic unit. The words are separated by space. The words are extracted from the document and are further processed into phrases, concepts. The words, phrases, and concepts are described as follows.

The word is the simplest feature item. Each feature item in the feature vector is associated with a word in the document. In general, case differences are ignored. When selecting words as feature items, the stop words are discarded, such as preposition and conjunction. In order to avoid the words from the same etymology appear many times, the stem of the words is extracted to represent the words have the same stem. For instance, “teach”, “teaching”, and “teacher” are converted as “teach” to be stored in the feature vector.

Using phrase as the feature item can overcome the weakness that simple words cannot reflect grammatical structure, the order in the paragraph, sentence, and words. Thus, an amount of information in the original document is not been effectively expressed. The meaning of words is often different in different phrases. The purpose of using phrases as feature words is to retain more information to distinguish document types. There are two methods to extract phrases from document. One is statistical method, which realizes phrase discovery through the co-occurrence probability statistics of words. This method can be suitable for a wide range of fields, but it needs a lot of training samples. The other is rule-based method, which identifies phrases through tagging dictionaries and word formation rules. However, this method is not flexible in grammar, and it is difficult to solve the ambiguity of word meaning. The dictionary cannot contain all the natural language phrases. Thus, it is difficult to exhaust all the rules of words.

Using concept as the feature item maximizes the internal similarity and minimizes the similarity between classes by combining the words in the source document according to a certain relationship. And then, the words are abstracted to the concept level to generate feature items. The feature items generated in this way contain more semantic information and have lower redundancy because similar information is merged.

In addition, some forms can also be used to generate feature items, such as tuple, some regular pattern. However, words, phrases, and concepts are most widely used form in document representation. Both phrases and concepts can be regarded as the combination or synthesis of the words. This paper adopts words as feature items in document representation.

Feature selection in document representation refers to discard the words that cannot contribute or have little contribute to distinguish document type. The feature selection can reduce the computational complexity on the basis of text preprocessing. There are several principles to select feature items. First, we should select those language units that contain more semantic information and have stronger ability to express the text as feature items. Second, the distribution of the text on these feature items should have obvious statistical regularity. Third, the selection process should be easy to implement and the associated time and space complexity.

The word can express whole semantic information. However, not all words are suitable as feature items. High frequency words and low frequency words are less effective than medium frequency words. The reason is that high frequency words in all articles have similar high frequency, low frequency words appear less in the text. Both high frequency word and low frequency word are not suitable to analysis by using statistical methods. The medium frequency words are most relevant to the topic of the document.

The common used feature selection for text classification includes text frequency, information Gain, mutual information, χ^2 -test, and term strength. It is important to choose the best feature selection among different feature selection algorithms. The performance of the vector space model directly depends on the selection of feature items and the calculation of weight.

The classical weight of feature item must considers term frequency and inverse document frequency (Havrlant 2017). The term frequency refers to the number of the words appearing in the document. The inverse document frequency refers to the quantitation of the distribution of words in the document. The common method to calculate inverse document frequency is represented as

$\log_2\left(\frac{N}{n_k} + 0.01\right)$. The N represents the number of documents in the document set, while n_k represents the number of the documents that contain the word.

The weight can be represented by combining term frequency and inverse document frequency as follows:

$$w_{i,k} = tf_{i,k} \times \log_2\left(\frac{N}{n_i} + 0.1\right) \quad (7)$$

In Equation (7), $tf_{i,k}$ represents the term frequency that word W_k occurs in document D_i , $w_{i,k}$ represents the weight of word W_k in document D_i ($k = 1, \dots, m$, m is the number of words). The vector is normalized by using the following equation:

$$w_{i,k} = \frac{tf(W_k, D_i) \times \log_2\left(\frac{N}{n_i} + 0.1\right)}{\sqrt{\sum_{W_k \in D_i} \left(tf(W_k, D_i) \times \log_2\left(\frac{N}{n_i} + 0.1\right) \right)^2}} \quad (8)$$

The above formula is based on the assumption that the most meaningful feature words to distinguish documents should be those words that appear frequently enough in documents and less frequently enough in other documents in the document set. An improvement of the weight is represented as follows:

$$w_{i,k} = N(W_k, D_i) * \left(\log_2 \frac{N(W_k)}{N} \right)^2 \quad (9)$$

In Equation (9), $N(W_k, D_i)$ is the times that word W_k occurs in the document D_i , $N(W_k)$ is the times that word W_k occurs in the training corpus, N the sum of the times that all words occur in the training corpus. The extracted features are used to represent the document. Then, the training corpus consisting of extracted features are used to learn a classification model. The common used classification model include: k nearest neighbor (Gou 2019), Gaussian processing (Manogaran 2018), logistic regression (Ranganathan 2017), and support vector machine (Cervantes 2020) etc. Since the processing to learn support vector machine is time-consuming, this paper adopts reduced support vector machine (Zhu 2017) as the classification model.

4. DOCUMENT TYPE RECOGNITION BY USING CLASSIFIER

Let $X \times Y$ represent the training set. The X is the feature set and Y is the label set. The $x_i \in \mathbb{R}^n$ is the feature of the i^{th} document, y_i is the associated label. In classical support vector machine $y_i \in \{+1, -1\}$. The aim of support vector machine is to find an optimal hyperplane $f(x) = w^T \varphi(x) + b$ via maximize the minimum margin between positive class and negative class. The optimal hyperplane is obtained by the following programming:

$$\begin{aligned} \min_{w, \rho} \quad & \frac{1}{2} w^T w + C \sum_{i=1}^l \xi_i \\ \text{s.t.} \quad & y_i (w^T \varphi(x_i) + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, l \\ & \xi_i \geq 0, \quad i = 1, 2, \dots, l. \end{aligned} \quad (10)$$

In Equation (10), l represents the number of samples in the training set, C is the penalty factor to balance the risk error and structural error, $\phi(x_i)$ is the image of sample x_i in the kernel reproducing Hilbert space to treat non-linear separable classification problem. The $\phi(x_i)$ is an implicit function, whose inner product can be computed via $\langle \phi(x_i), \phi(x_j) \rangle = k(x_i, x_j)$. The kernel function is an explicit function which can be computed directly. The common used kernel functions include: linear kernel, polynomial kernel, Gaussian kernel, exponential kernel etc. When the dataset is linear separable, the linear function is adopted.

The Equation (10) is a convex programming whose solution is the same as that of its dual programming. After introducing the Lagrange multiplier α_i for the constraints $y_i (w^T \varphi(x_i) + b) \geq 1 - \xi_i$ in the Equation (10), the dual form of Equation (10) is written as follows:

$$\begin{aligned}
& \min_{\alpha} \quad \alpha^T Q \alpha - \alpha \\
& s.t. \quad \sum_{i=1}^l y_i \alpha_i = 0, \quad i = 1, \dots, l \\
& \quad \quad 0 \leq \alpha_i \leq C, \quad i = 1, \dots, l
\end{aligned} \tag{11}$$

In Equation (11), α is the vector form of Lagrange multipliers, Q is the kernel matrix whose element is $Q(i, j) = y_i y_j k(x_i, x_j)$. The Equation (11) can be solve be decomposition algorithm or chunking algorithm. The weight and bias of the optimal hyperplane are rewritten as follows:

$$w = \sum_{x_i \in SVs} \alpha_i \varphi(x_i) \tag{12}$$

$$b = y_i - \sum_{x_j \in SVs} \alpha_j k(x_i, x_j), \quad 0 < \alpha_i < C \tag{13}$$

In Equation (12) and (13), the SVs represents the support vector set. The support vector is the sample which is with non-zero Lagrange multiplier ($0 < \alpha_i \leq C$). Then, the optimal hyperplane is rewritten as follows:

$$f(x) = \sum_{x_i \in SVs} \alpha_i k(x, x_i) + b \tag{14}$$

The Equation (14) is only determined by the support vectors. The complexity of Equation (14) is determined by the number of support vectors. The scale of programming (13) and the number of support vectors are related with the size of the training set. When the size of training set is huge, the number of support vectors is also very large. Thus, it is time-consuming to solve programming (13). In order to solve this issue, the sample reduction strategy is introduced into support vector machine model learning.

The optimal hyperplane $f(x)$ is only determined by minor support vectors. If removing the samples that would not become support vectors, the result of support vector machine will not change. Let $\mathbf{X}' \times \mathbf{Y}'$ be the set consisting of all potential support vectors. The reduced support vector machine learning on reduced training set is formulated as follows:

$$\begin{aligned}
& \min_{w, \rho} \quad \frac{1}{2} w^T w + C \sum_{i=1}^l \xi_i \\
& s.t. \quad y_i (w^T \varphi(x_i) + b) \geq 1 - \xi_i, \quad \{x_i, y_i\} \in \mathbf{X}' \times \mathbf{Y}' \\
& \quad \quad \xi_i \geq 0, \quad i = 1, 2, \dots, l.
\end{aligned} \tag{15}$$

The dual form of programming (15) is the same as that of programming (11). The scale of programming (15) is related with the size of $\mathbf{X}'$. Let l' represent the size of set $\mathbf{X}'$. In general, $l' < l$. The potential support vectors generally locate in the boundary region of each class or near the optimal hyperplane. This paper adopt the distribution of nearest neighbors (Zhu 2014) to detect the samples in the boundary region of each class as the candidate support vectors for reduced support vector machine learning. The distribution of nearest neighbors is described by the following equation:

$$d_{x_i} = \sum_{x_j \in NN_k(x_i, X_{y_i})} \frac{\langle x_i, x_i \rangle - \langle x_i, x_j \rangle - \frac{1}{k} \sum_{x_p \in NN_k(x_i, X_{y_i})} \langle x_i, x_p \rangle + \frac{1}{k} \sum_{x_p \in NN_k(x_i, X_{y_i})} \langle x_j, x_p \rangle}{\sqrt{\langle x_i, x_i \rangle - \frac{2}{k} \sum_{x_p \in NN_k(x_i, X_{y_i})} \langle x_i, x_p \rangle + \frac{1}{k} \sum_{x_p, x_q \in NN_k(x_i, X_{y_i})} \langle x_p, x_q \rangle} \sqrt{\langle x_i, x_i \rangle - 2\langle x_i, x_j \rangle + 2\langle x_j, x_j \rangle}} \quad (16)$$

In Equation (16), $NN_k(x_i, X_{y_i})$ represents nearest neighbor set of sample x_i in set X_{y_i} , the set X_{y_i} represents the set consists of all samples with same class of sample x_i . In general, the Equation (16) is close to 1 if the sample locates in the boundary region of a class, and close to 0 if the sample locates in the interior of a class. Then, we only need to retain the samples with high values as the candidate training set for reduced support vector learning. The whole procedure is summarized as shown in Algorithm 1.

In Step 3 of reduced support vector machine, l_p represents the number of samples in positive class. In Step 6 of reduced support vector machine, l_n represents the number of samples in negative class. The parameter δ represents the size of retained subset of the training set. The larger δ is, the more samples are retained in reduced support vector machine.

5. EXPERIMENTS AND SIMULATIONS

In this section, we use the framework in Figure 1 to recognize the document type. We adopts the documents from Reuter's news. The Reuter's news is a standard dataset which is widely used in the research of text classification. The documents are denoted manually and processed as the fixed form. The Reuter's news include four types: corporate/industrial (CCAT), government/social (GCAT), markets (MCAT), and economics (ECAT). The corporate/industrial includes strategy/plans, legal/judicial, and share listing. The government/social includes sports, environment and natural word. The markets include equity markets, bond markets. The economics includes economics performance, monetary/economic. Each document type contains 600 documents and there are 2,400 documents in total for training. Each document is represented as the extracted features. The classical support vector machine only can deal with binary class classification problem. In order to address multi-class classification problem, the one versus one strategy is adopted. In one versus one strategy, 6 binary class classification models are learnt, include CCAT versus GCAT, CCAT versus MCAT, CCAT versus ECAT, GCAT versus MCAT, GCAT versus ECAT, and MCAT versus ECAT. The learnt model is evaluated on a test set which consist of 800 documents in total, 200 document per type.

The document type recognition framework is evaluated from validity, computational complexity and simplicity. The validity measures the ability whether the framework can classify document type

Algorithm 1. Reduced support vector machine

- Step 1:** calculating values of Equation (16) for all samples in positive class;
Step 2: sorting the samples in positive class according to the values of Equation (16) in descending order;
Step 3: retaining the top $\delta * l_p$ samples in positive class to construct X'_p ;
Step 4: calculating values of Equation (16) for all samples in negative class;
Step 5: sorting the samples in negative class according to the values of Equation (16) in descending order;
Step 6: retaining the top $\delta * l_n$ samples in positive class to construct X'_n ;
Step 7: combing X'_p and X'_n as X' and learning support vector machine model on $X' \times Y'$.

correctly. The computational complexity includes time complexity and space complexity. The time complexity includes training time and test time. The simplicity requires the classification model to be as simple as possible. The validity is the most important index. If a document type recognition model has simple structure, is easy to be learnt, but cannot classify document correctly, the document type recognition is useless. The validity is evaluated by precision, recall, and F1-measure.

Let TP, FP, TN, and FN represent true positive, false positive, true positive, and true negative, respectively. Then, precision, recall, and F1-measure are defined as following equations:

$$Precision = \frac{TP}{TP + FP} \quad (17)$$

$$Recall = \frac{TP}{TP + FN} \quad (18)$$

$$F_1 = \frac{2Precision * Recall}{Precision + Recall} \quad (19)$$

The experimental results are reported in Table 1 in terms of precision, recall, and F1-measure. The features are selected by Boolean model, vector space model, probability model, and term based model. The classification algorithm adopts support vector machine (SVM) and reduced support vector machine (RSVM).

From the results in Table 1, when support vector machine is used as classification algorithm, the term-based model achieves 80.73%, 82.35%, 81.89% for precision, recall, and F1-measure. The results is higher than Boolean model 9.5%, 8.83%, and 9.22%; is higher than vector space model 6.2%, 6.99%, 7.96%; and is higher than 6.34%, 6.94%, 7.02% for precision, recall, and F1-measure. The reduced support vector machine retains 15% of the training set. The difference between support vector machine and reduced support vector machine is less than 0.15% for precision, recall, and F1-measure.

The complex of classification model of support vector machine and reduced support vector machine is reported in Table 2 in terms of training time and the number of support vectors. The complex of support vector machine model depends on the number of support vectors. The number of support vectors in Table 2 is the average of the number of the support vectors of 6 binary classification models. The reduced support vector machine retains 15% of the training set.

From the results in Table 2, it can be found that the reduced support vector consumes 6.9% of the support vector machine and the support vectors of reduced support vector machine is only 16.15% of that of support vector machine.

In Figure 2, we report the results of reduced support vector machine with different percentage of the training set. The size of retained subset ranges 5%, 10%, 15%, 20%, and 25%. The document is represented by termed based model.

From the result of Figure 2, it can be found that when 15% of the training set is retained, the reduced support vector machine can maintain the performance which is very close to that of support vector machine.

Figure 3 reports the confusion matrix when 15% of the training set is retained.

From the result of Figure 3, it can be found that the accuracy can achieve 78%, 80.5%, 84.5%, 79.5% for CCAT, GCAT, MCAT, and ECAT, respectively.

Table 1. The performance comparison of document type recognition

		SVM	RSVM
Boolean model	Precision (%)	71.23	71.09
	Recall (%)	73.52	73.39
	F1-measure (%)	72.67	72.53
Vector space model	Precision (%)	74.53	74.47
	Recall (%)	75.36	75.42
	F1-measure (%)	73.93	74.03
Probability model	Precision (%)	74.39	74.41
	Recall (%)	75.41	75.32
	F1-measure (%)	74.87	74.73
Term-based model	Precision (%)	80.73	80.59
	Recall (%)	82.35	82.42
	F1-measure (%)	81.89	81.83

Table 2. The model complexity comparison between support vector machine and reduced support vector machine

	SVM	RSVM
Training time (sec.)	252.31	17.42
No. of SVs	87	14

Figure 2. The relation between the size of the retained subset in reduced support vector machine and precision, recall, and F1-measure

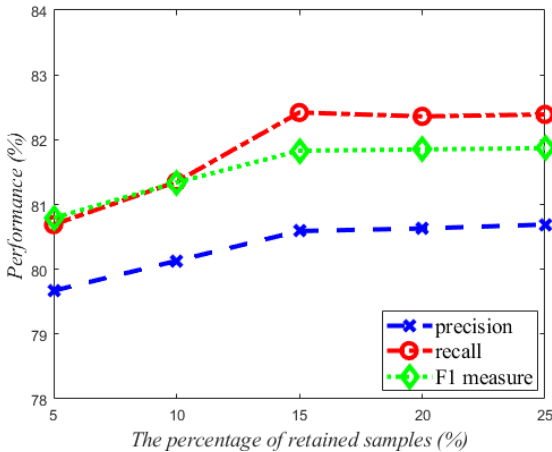


Figure 3. The confusion matrix of document type recognition of reduced support vector machine

	CCAT	GCAT	MCAT	ECAT	Accuracy (%)
CCAT	156	23	23	29	67.53
GCAT	13	161	5	7	86.56
MCAT	27	7	169	5	81.25
ECAT	4	9	3	159	90.86
Accuracy (%)	78.00	80.50	84.50	79.50	80.63

6. CONCLUSION

The automatical document type recognition and analysis plays an important role in the massive emerging web information. From, the web documents are collected and denoted the type by domain experts. The collected documents are constructed as training set to learn a classification model. Then, the future document is represented as term-based features and the features are input into classification model to determine the document type. In the proposed document type recognition framework, the term-based model is adopted as feature representation method. The reduced support vector machine is adopted as classification algorithm. In order to solve multi-class classification problem, the one versus one strategy is adopted. The results of the experiments on Reuter's news dataset show that most of the document type can be correctly identified by the proposed document type recognition framework.

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Guest Editorial Preface

Special Issue on Intelligent Data Mining and Engineering-Assisted Mobile Software System Analysis and Design

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In the past decades, mobile systems become more and more popular. It is critical to design associated software systems in this mobile era. Nowadays, various software systems with intelligent functions cooperatively provide people-oriented services by massive intelligent software models and technologies (e.g., machine learning approaches and complex network techniques), network paradigms (e.g., edge computing, fog computing and cloud computing) and smart hardware devices (e.g., IoT devices, wearable devices, and smart sensors). It is worth noting that the current analysis and design of software systems concentrate on two nature attributes: complexity and sustainability. On one hand, since heterogeneous types of sensors spring up, the sensed and interfaced data in software systems are more difficult to be mined and explored, which induces spectacular complexities in data collecting, saving, processing, and transmitting. On the other hand, sustainable solution is a nontrivial goal for any software system. For instance, the internet of things (IoT)-based smart agriculture technology has greatly reduced agriculture's dependence on labour.

Although an enormous amount of effort has been made, the intricate and dynamic data analysis in software systems still requires multi-scale and diverse intelligent data mining and engineering approaches to support. It is urgently intractable that how to leverage intelligent data mining and engineering approaches to satisfy the demands of a software system which is high complexity and sustainability. This primary intent of this special issue is to attract and solicit the recent advances from academic researchers and industry practitioners in the development and improvement of software systems by adopting data mining and data engineering technologies. The eight papers in this special issue cover the theme of this special issue.

The first article, "Intelligent Data Mining-Based Method for Efficient English Teaching and Cultural Analysis," by Qing Ai and Hongyu Guo mainly studies an intelligent English teaching method to improve the quality of English teaching. Specifically, the random forest is firstly used to analyse and excavate the grammatical and syntactic features of the English text. Then, the decision tree-based method is proposed to make a prediction about the English text in terms of its grammar or syntax issues.

The second article, "Intelligent Skiing Posture Detection and Recognition Through Internet of Bodies," by Peihua Liu establishes a system to automatically recognize the skiing posture which can help athletes grasp the skiing postures. First, the skiing images are collected by distributed camera. Second, the skeleton features are extracted to learn a classification model which is used to recognize and adjust skiing postures. Lastly, the analytical results of posture recognition is returned to athletes through Internet of bodies.

The third article, "Mobile Edge Computing to Assist the Online Ideological and Political Education," by Dan Wang and Jian Zhao utilizes a combined optimization to minimize total cost and maximize QoE simultaneously to provide users with better quality of experience (QoE) in the mobile video business.

The fourth article, “Online Music Style Recognition via Mobile Computing,” by Lizhu Yuan and Yue Zhang adopts machine learning technology to establish an automatic music style recognition system. First, the online music is process by waveform analysis to remove the noises. Second, the denoised music signals are represented as sample entropy features by using empirical model decomposition. Lastly, the extracted features are used to learn a relative margin support vector machine model to predict future music style.

The fifth article, “Prediction of Football Match Results Based on Edge Computing and Machine Learning Technology,” by Yunfei Li and Yubin Hong proposes a football match result prediction method based on edge computing and machine learning technology. Specifically, it first extracts some game data from the results of the previous games to construct the common features and characteristic features, respectively. Then, the feature extraction and classification task are deployed to multiple edge nodes. Finally, the results in all the edge nodes are uploaded to the cloud server and fused to make a decision.

The sixth article, “Video Sequence Analysis for on Table Tennis Player Ranking and Analysis,” by Xiaoni Wei establishes a table tennis players’ ranking model by using competition videos and prestige scores in the competition relationship network. The competition relationship network is constructed according to the competition events between athletes.

The seventh article, “English Article Style Recognition and Matching by Using Web Semantics,” by Mi Zhou and Lina Peng adopts termed-based model to extract the features in web semantics to represent document. The extracted web semantics features are used to learn a reduced support vector machine which help people automatically classify text and liberates them from the tedious document processing work.

The eighth article, “Aesthetic Evaluation of Interior Design Based on Visual Feature,” by Zhen Zhang and Jianwei Ban implements an intelligent aesthetic evaluation of interior design framework by using computer vision and artificial intelligence technologies, which can help people choose the appropriate and effective interior design from collected images or mobile digital devices.

In conclusion, the articles presented in this special issue demonstrates fruitful research in the field of intelligent data mining and engineering in mobile software system analysis and design. We wish to thank both the authors and the reviewers for their hard work in helping us assemble this Special Issue, and would also like to express our sincere gratitude to the Editor-in-Chief, Dr. Agustinus Waluyo, for providing this opportunity and lots of guidance throughout the process.

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Intelligent Data Mining-Based Method for Efficient English Teaching and Cultural Analysis

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ABSTRACT

The emergence of online education helps improve the traditional English teaching quality greatly. However, it only moves the teaching process from offline to online, which does not really change the essence of traditional English teaching. In this work, the authors study an intelligent English teaching method to further improve the quality of English teaching. Specifically, the random forest is firstly used to analyze and excavate the grammatical and syntactic features of the English text. Then, the decision tree-based method is proposed to make a prediction about the English text in terms of its grammar or syntax issues. The evaluation results indicate that the proposed method can effectively improve the accuracy of English grammar or syntax recognition.

KEYWORDS

Data Mining, Decision Tree, English Teaching, Intelligent, Random Forest

1. INTRODUCTION

Since the beginning of the 21st century, the world economy and arms competition have become increasingly fierce, and the demand for English professionals has gradually expanded. In order to meet the needs of rapid economic development, different colleges and universities have implemented a series of reforms in view of the existing English teaching models (Guan 2018). Among them, English grammar and syntactic analysis are the key points in the foundation, and they are the necessary conditions for the establishment and cultivation of compound English talents. With the introduction of the compound talent training mode of “English + news”, colleges and universities gradually begin to actively explore diversified English teaching models, give full play to students’ practical and innovative abilities, and carry out comprehensive practical education around the creative ability of graduates (Mahbub 2021). It plays an important role in cultivating the new ability of English majors.

At present, although the level of English teaching in colleges and universities in China is constantly improving, English, as a public subject, has not been deeply integrated with specific fields and vocational skills, which is reflected in the unfamiliarity of English grammar rules in many fields (Albiladi 2019). French semantic logic is not clear and so on. Therefore, in order to fully meet the needs of social development and educational reform in colleges and universities, we should actively explore an English teaching model dominated by the cultivation of students’ professional ability, focusing on the cultivation of basic English analysis and judgment, and realizing the simultaneous development of students’ English proficiency and vocational skills plays an important role in improving students’ English ability (Mirbabayeva 2020).

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In the context of the current big data era, the rapid development of technology has also brought opportunities for the development of college English teaching. Especially in the current teaching, the application of some advanced network technology and information processing technology has greatly facilitated English teaching, changed the traditional English teaching mode, made teaching more diversified and more interesting, and significantly improved students' interest in learning (Elyas 2018). With the assistance of big data, school institutions can build an effective English grammar and syntax database, record students' phased English learning, and then organize and analyze them as a basis to constantly optimize teaching ideas, improve the pertinence of English teaching, so as to fully meet the diversified and dynamic needs of students (Hudson 2005). Under the application of big data, we can also strengthen the dynamic monitoring of teaching quality and improve the timeliness of teaching. In addition, the application of information technology can help students understand knowledge and enrich classroom content, so as to improve students' learning efficiency and teaching quality (Guo 2021). Therefore, in order to meet the needs of the development of major industries, colleges and universities at home and abroad gradually combine data mining and other related contents to carry out a wide range of English teaching, including basic grammar, semantics and so on (Lou 2020).

However, big data brings both opportunities and challenges to English teaching. The challenges are mainly reflected in two aspects. First of all, the massive teaching resource in big data's era have weakened the English teaching model based on teaching materials. In the current efficient English teaching system, due to the solidification of the teaching materials used, with the continuous reform and development of teaching, it may have been unable to meet the needs of the current development (Weng 2020). Moreover, a large amount of English teaching resource are usually discrete and there are a lot of repetitive content, which makes it very difficult to obtain effective English knowledge (Bienkowski 2012). Therefore, although big data is rich in resource, students have a negative attitude towards finding effective resource from the vast amount of data on the Internet, so it is difficult to improve their interest in learning. In addition, the teacher-led classroom learning model leads to students' passive acceptance of knowledge and lack of interaction. In the long run, it will not only lead to professional exhaustion of teachers, but also can not bring effective feedback to students' English learning effect (Mihaescu 2021).

Based on the above consideration, in order to improve the quality of English teaching, this paper explores the important writing features of English sentences at the grammatical and syntactic level by using random forest and logical regression machine learning algorithms. Based on this feature, the decision tree model is used to optimize and predict and judge the grammatical and syntactic correctness of sentences, so as to improve the quality of English teaching. The purpose of this paper is to provide a basis for improving English teaching management and improving the quality of English teaching. The main contributions of this paper are as follows:

- By using the random forest theory, he fully analyzes and excavates the grammatical and syntactic features of the text in English teaching, and then extracts the English sentence features at the grammatical and syntactic levels, which can be used for further modeling and analysis of English grammar teaching.
- According to the extracted effective sentence features, an English sentence classification method based on decision tree method is proposed, and an English grammar prediction model is established by using a large amount of English grammar teaching data for learning and classification, and then provide the model and method basis for efficient and automatic English teaching.
- By using a large number of data for statistical analysis and theoretical verification of the proposed model method, the results show that the proposed method can effectively improve the accuracy of English grammar recognition, and then improve students' autonomous learning ability.

The organizational structure of this paper is as follows: the second chapter mainly introduces the related work. The third chapter introduces the model and method proposed in this paper. The fourth chapter introduces the research results of this paper. The fifth chapter is a summary.

2. OVERVIEW OF RELATED TECHNOLOGY AND RESEARCH

Based on the existing research, we divide the related work into two categories, namely, the traditional English teaching research based on data mining and the English teaching research based on intelligent data mining. The following first summarizes the related technologies and current situation of the two kinds of research work, and then carries on the summary and discussion.

2.1 Traditional Data Mining Based English Teaching Work

With the rapid development of information technology, a large amount of data has been accumulated in the daily operation and work in the field of English teaching. With the help of these data, relevant personnel can use artificial or intelligent methods to deal with related work and find many rules and patterns (Dogan 2020). Through the analysis of these data, we can provide important information for the decision-making of English teaching quality improvement and mode reform, which is helpful to improve work efficiency. Data mining technology is an interdisciplinary subject which integrates artificial intelligence technology, visualization technology, database technology, statistics and machine science. It was first proposed in 1989 (Romero 2020). At present, people have different definitions of data mining, among which it is generally accepted that data mining is from a lot of incomplete, large, noisy, fuzzy and random data, through related programs to extract some hidden information and knowledge that people do not know (Kumar 2020). For English teaching, the process of using data mining technology also includes three stages, namely, English text data preparation, English rule feature mining and result expression.

The first is the English text data preparation stage, which mainly covers three sub-steps: text data acquisition, selection, data pre-processing and data conversion. The acquisition of English text data usually adopts the way of web crawler, and the selection is made according to the main goals and tasks of data mining of relevant English institutions. Mining some analytical data from the original and existing data to form the target data of data mining, which lays a good foundation for later data mining (Taleb 2015). Data pre-processing is to make the target data meet the requirements of data mining through the analysis and processing of the target data, which is also the specific function of data mining. In this link, relevant personnel are needed to calculate the missing values of English text data, so as to ensure the standardization of the data format. The second is to mine the grammar rules and rules in the English text, and we need to choose the appropriate mining methods according to the actual goals and requirements, such as decision tree, support vector machine and so on (Zhang 2020, Wen 2020). For example, Zhang W. et al. use the decision tree method to mine English syntactic rules (Zhang 2020), while Wen H. et al. use the support vector machine algorithm. In comparison, the support vector machine method is more suitable for complex English grammar and grammar situations (Wen 2020), while the decision tree method has the advantages of high efficiency and low cost. Finally, there is the expression of the results of English text mining, and the rules to be mined are expressed in a general form.

2.2 Intelligent Data Mining Based English Teaching Work

The traditional methods of machine learning in the process of English teaching based on data mining can be divided into two categories, namely, supervised learning and unsupervised learning. The main difference between the two models lies in whether the data used in the English teaching model not only has the basic characteristics, but also has the corresponding label value to classify it. Generally speaking, there are two main kinds of supervised learning problems, classification and regression. The main algorithms of supervised learning include K-proximity analysis, linear model, naive Bayesian

classifier, decision tree, kernel support vector machine, neural network and so on. There are two common types of unsupervised learning: data set transformation and clustering. The unsupervised transformation of data sets is an algorithm for creating new representations of data. Compared with the original representation of data, the new representation may be more easily understood by people or other machine learning algorithms. The clustering algorithm divides the data into different groups, each group contains similar items, and the main clustering algorithms are K-means clustering and agglomeration clustering (Liu 2021). Classification is to label the data according to a certain standard, and then classify them according to the label. Clustering is a process of finding out the reasons for the clustering of data through analysis without a “label” in advance (Zhen 2021).

Many scholars at home and abroad have used intelligent data mining technology to study English teaching and obtained more research results. For example, Xu Z. et al. use stepwise regression and neural network techniques to analyze college students' English learning achievements and their influencing factors (Xu 2019). Pangaribuan T. et al. analyzed a large number of learning data generated in the process of online English learning with the help of statistical analysis and visualization, association rule algorithm and clustering algorithm (Pangaribuan 2018). According to the results of the analysis, some thoughts and suggestions on the supervision and management of web-based English learning process are given. Cui S. et al. applied the multiple linear regression model to predict the academic performance of students in hybrid college English courses based on physical classroom and cloud learning platform, and carried out teaching intervention to improve their learning effect according to the prediction (Cui 2017). Dalton-Puffer C. et al. constructed a classified prediction model of online English learners' academic performance by using the method of nested integrated learning, which provides a reference for the research on the influencing factors and predictive modeling of online English learners' academic performance, and is also helpful to the practice of online learning academic early warning, academic performance prediction and evaluation (Dalton 2011). Eramona F. et al. applied the clustering algorithm to the analysis of English-related course scores and learning information of undergraduate students in online academic education, and realized the subdivision and prediction of adult degree English examination scores (Eramona 2014).

2.3 Discussion

By collating the results of domestic research on the application of data mining technology to English teaching, it is found that many studies have relatively focused on the traditional English teaching model, that is, the use of simple data statistical methods to mine and improve the quality of English teaching. Although the emergence of online education is a great progress in the English teaching model, it only moves the process of data statistical analysis from offline to online, from manual statistics to machine statistics, which has not changed in essence. Although the application of machine statistical analysis is larger and more efficient, it can not flexibly adapt to various situations in English teaching. Therefore, this paper mainly studies intelligent English teaching methods by combining data mining technology and related machine learning technology, so as to provide a basis for improving the quality of English teaching.

3. INTELLIGENT DATA MINING METHOD FOR ENGLISH TEACHING AND CULTURAL ANALYSIS

3.1 Data structure Analysis and Data Pre-Processing in English Teaching

In order to apply intelligent data mining technology to English teaching, it is necessary to analyze the syntactic and grammatical complexity in the process of English teaching. Specifically, syntactic complexity refers to the scope and complexity of different forms of language output, which is mainly used to evaluate language proficiency, describe language ability and measure language development. It is an important index to evaluate learners' language development. At present, the research on syntactic

complexity at home and abroad mainly focuses on: (1) horizontal research, that is, by comparing English data representing different English learning levels or writing quality, explore the measurement indicators of syntactic complexity that can effectively distinguish different English learning levels or writing quality; (2) Longitudinal study, that is, by analyzing the multiple writing corpus of the same group of learners at different time points, to explore the development model of syntactic complexity. However, researchers pay less attention to the correlation between the measured syntactic complexity of English learners' writing texts and their writing scores.

In order to solve the bottleneck of data analysis in the study of grammatical and syntactic complexity in English teaching, in order to help researchers carry out a more effective study of English syntactic complexity, five major categories (14 subcategories) are used to analyze the syntactic complexity of written English texts, as shown in Table 1. Specifically, it includes: 1) English sentence unit length (Mean Length of Sentence (MLS), Mean Length of T-unit (MLT), Mean Length of Clause (MLC)); 2) sentence complexity (Clause per Sentence (CS)); 3) subordinate sentence usage (Clause per T-unit (CT), Clause per T-unit Ratio (CTR), Dependent Clause per Clause (DCC), Dependent Clause per T-unit (DCT)). 4) the amount of coordinate structure (Coordinate Phrase per Clause (CPC), Coordinate Phrase per T-unit (CPT), T-unit per Sentence (TS)), and 5) specific phrase structures (Complex Nominal per Clause (CNC), Complex Nominal per T-unit (CNT), Verb Phrase per T-unit (VPT)).

Now, using the web crawlers and other technical means to obtain a large number of English teaching data from open source websites, however, these data need to be pre-processed before they are used. Data pre-processing is the preparation work before data mining, which aims to provide standardized and targeted data for data mining, reduce the data processing capacity of data mining algorithms, improve mining efficiency, and finally improve the accuracy of the model. In this paper, the pre-processing operation is mainly concerned with the standardization and unity of the data format. The complexity of English syntax is different, which may lead to a large gap in the range of corresponding eigenvalues, and the lack of relevant pre-processing will affect the final result. Therefore, in this paper, the complexity data of English text is scaled by a certain proportion, and the maximum-minimum normalization method is used to map the value to [0, 1] interval, that is, given the complexity value x of different English syntactics. then the corresponding normalization pre-processing process is defined as follows:

$$x_{new} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where $x_{\min}$ is the minimum value; $x_{\max}$ is the maximum value; x_{new} is the new value of x after the operation of normalization.

3.2 Model Establishment

Based on the English sentence data obtained by pre-processing, this paper mainly uses random forest to construct English sentence data classifier. For each decision tree in the random forest model, the training set they use is sampled back from the total training set, which means that some samples in the total training set may appear in the training set of a tree many times. or it may never appear in the training set of a tree. When training the nodes of each tree, the features used are randomly extracted from all the features according to a certain proportion. Then, assuming that the overall feature of the English text is M , we can set such proportion as $\sqrt{M}$.

The overall training process of this model can be described as follows. First of all, we calculate the overall English text data satisfying the format and denote it by S . Then, the testing data set and the feature dimension can be denoted by T and F respectively. After that, according to the model

that need to be trained, we should also decide the training parameters which include the number of decision trees (denoted by t) that consist of the random forest, the length of each tree (denoted by d), and the number of features of each node of the tree (denoted by f). Then, for any decision tree, we define the information entropy for it, as follows:

$$Ent(S) = -\sum_{k=1}^f p_k \log_2 p_k \quad (2)$$

where p_k is the rate between the k -th type data sample and the overall data set S . Then, the objective is to minimize $Ent(S)$, since the lower the value of $Ent(S)$, the better the training performance.

Then, extracting the training set $S(i)$ from S as the training samples for the root node which is the starting node for the overall training process. However, we should be aware that the training process need to be terminated by certain condition which is defined as follows:

$$Gain(S) = Ent(S) - \sum_{i=1}^F \frac{|S^i|}{|S|} Ent(S^i) \quad (3)$$

That is, if the value of the gain value $Gain(S)$ cannot be reduced anymore, the training process for this node will be terminated. After that, the current node will be regarded as the leaf node. Despite this, if the termination condition is not reached, we randomly select the f dimensional features from the overall F dimensional features without putting them back. Leveraging the f dimensional features, we then search for the feature (assume it is indexed by k) with the best classification results and its threshold denoted by *threshold*. Then, the samples on the current node with their k -th feature smaller than *threshold* will be separated to the left node, while the rest will be separated to the right node. Now, repeating the process until all the nodes are trained or marked as leaf node. Once all the nodes are trained, then the tree is established.

3.3 Prediction Based on Decision Tree

Now, we have well trained the model, based on which we can next use it to prediction the syntax issue of any English text to improve the English teaching quality. Given any English text with fixed structure, we first pick one decision tree from the random forest, and then the searching process starts from the root node of this decision tree. For the current node, if the value of *threshold* of this node is larger than the feature value of this English text, we go into the left node. Otherwise, we go into the right node. Repeating the above searching process until the leaf node is reached. Then, a result of this tree is output, that is, this English text has grammar or syntax issue or not, which can be defined by the following binary variable.

$$z_i = \begin{cases} 0, & \text{this sentence has no grammar issue} \\ 1, & \text{otherwise} \end{cases} \quad (4)$$

where $i > 0$ indicates the number of the decision tree and we should run the above process through each tree. Assuming that there are N decision trees in the random forest, then the prediction results can be described as follows:

$$\{z_1, z_2, z_3, \dots, z_N\} \quad (5)$$

To make it easy-to-be-understood, we generate a probability for English syntax or grammar issue prediction, that is,

$$p = \frac{\sum_{i=1}^{|N|} z_i}{|N|} \quad (6)$$

where $p \in [0,1]$ is the probability that the English text has grammar or syntax issue. Then, if

$$p > 1 - p \quad (7)$$

is true, we decide that this English text has grammar or syntax issue. Otherwise we judge there is no grammar or syntax issue.

The overall working process is shown in Fig. 1, as follows:

4. EXPERIMENTAL RESULTS

4.1 Setup

The key part of We use the crawler technology to obtain tremendous amount of English text data from online websites such as VIPKID and PALFISH. In order to make the model working efficiently, we select 100000 English text data, among which 80% of them are used as the training set and the rest 20% of them are used as the testing set.

To better test the result of the proposed method, we adopt three benchmarks as follows:

- Benchmark1 (Guo 2021): it uses the traditional statistical method to dig the regular from tremendous amount of English text data, such that we refer it as TSM in the comparison.
- Benchmark2 (Lou 2020): it uses the K-nearest neighbor method to explore the relationship between different English texts, such that we refer it as KNM in the comparison.
- Benchmark3 (Xu 2019): it relies on using the supported vector machine method to classify the categories of different English texts, such that we refer it as SVM in the comparison.

For convenience, the proposed method is referred to as Decision Tree based Method (DTM). Then, TSM, KNM, SVM and DTM are evaluated over the following metrics of English syntax statistical result, time overhead and prediction accuracy.

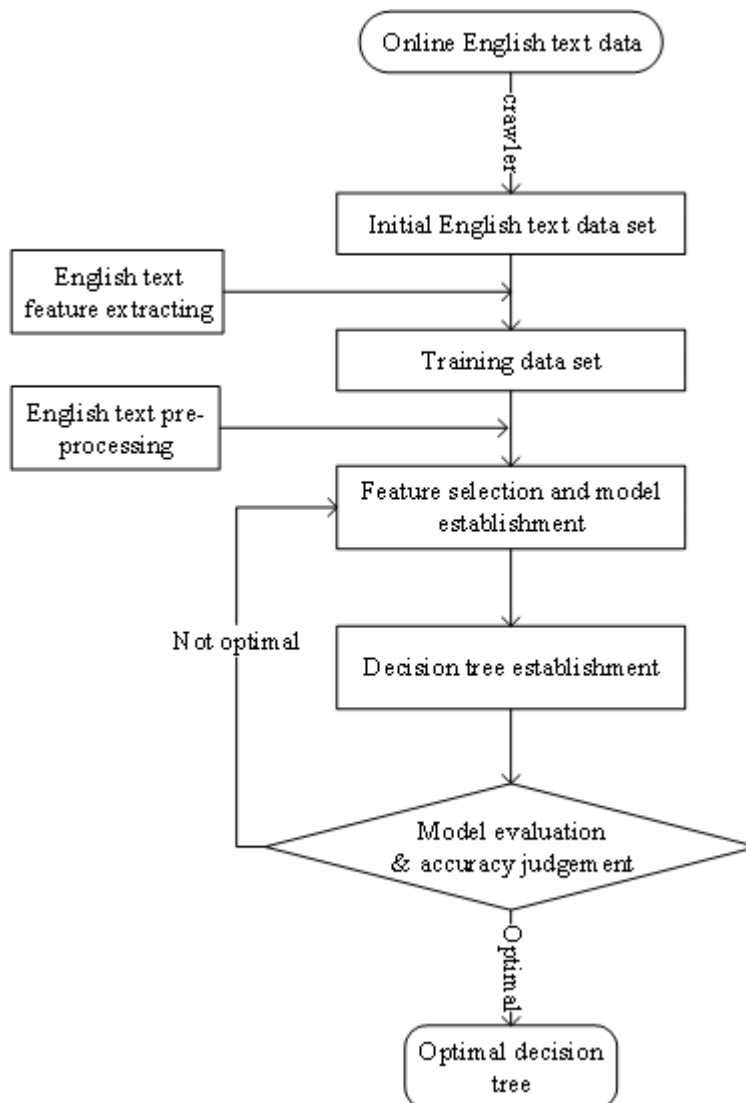
4.2 Results

4.2.1 English Syntax Statistical Result

As explained, we finally obtain 100000 English texts for training the English syntax model. In particular, the quality of such data is vital important for the prediction result. Hence, we first present the statistical results of the data set. Since the English text is classified into 14 kinds, the statistical results are summarized accordingly and shown in Table 2.

In particular, by observing the results, we can conclude that the obtained data can satisfy the requirements of the proposed method. On one hand, the diversity of the data can be guaranteed, since we have collected the data from 14 kinds of English texts and it is easy to note that the number of

Figure 1. The flow chart of the overall working process



each kind of English text exceeds zero. On the other hand, let us observe the distribution of these English text data, several phenomena can be found, that is: 1) the format of MLS of the English text has the largest mean value, while CPC has the smallest mean value; 2) MLT has the largest Max-value while CTT has the smallest Max-value; 3) the min value of MLS is the largest and that of DCC is the smallest; 4) the standard deviation of MLS is the largest and that of DCC is the smallest. This is reasonable, since MLS structure in English text is the most commonly used structure of the English text, while DCC is the not the general one.

4.2.2 Time Overhead

Now, we have the well prepared data for training the English syntax model. Next, we should start the training model. In this regard, the time overhead becomes a critical factor to evaluate the effectiveness

Table 2. Statistical results of the 100000 English texts

	Mean value	Max value	Min value	Standard deviation
MLS	35.12	391	5.35	39.66
MLT	27.21	392	6.55	33.25
MLC	12.38	115	1.785	9.14
CS	3.5	15	0.8	1.78
VPT	4.84	15	0.98	1.35
CT	2.12	9	0.78	0.95
DCC	1.41	2.441	0.02	0.1333
DCT	1.99	6.78	0.401	0.789
TS	1.07	5.12	0.048	0.31
CTT	0.556	1.1	0.037	0.22
CPT	0.489	4.2	0.02	0.364
CPC	0.255	2.57	0.02	0.178
CNT	3.52	15	0.24	1.398
CNC	1.198	5	0.25	0.385

of the proposed method. This metric is evaluated against the number of English texts and shown in Fig. 2.

Overall, by observing the training time achieved by the four methods, we can discover two obvious phenomena that: 1) DTM has the smallest time overhead and KNM has the second smallest time overhead, that is, given the same amount of training data, DTM spends the least time to finish training the corresponding model; 2)TSM has the highest time overhead, while SVM is the second highest. This is reasonable, because TSM relies on statistical technology which is highly proportional to the scale of the training data. As for DTM, despite the fact that its time overhead also increases with the increasing of the number of English syntax text, it increases slowly, due to the fact that the simple training structure of the decision tree. On the contrary, the training network structure of the supported vector machine is relatively complex for the purpose of achieving high prediction accuracy.

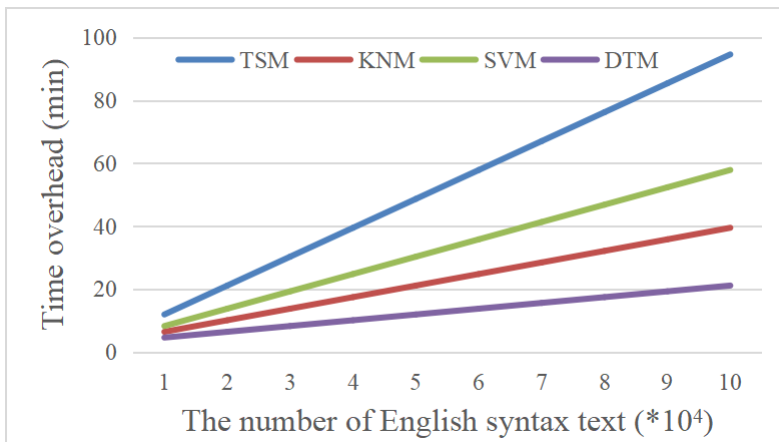
Another deep condition can be seen is that the time overhead of these four methods increases linearly when the number of the English syntax text is increasing, which is reasonable. On one hand, dealing with a lot of data requires time, which supports the situation that the time overhead increases with the increasing of the scale of data. On the other hand, the training model of these methods are fixed, that is, the network structure (i.e., how many nodes and how many links) and the training parameters used for model training are all known, such that the the linear relationship between them is fixed. Therefore, the increasing trend is also linear as shown in Fig. 2.

4.2.3 Prediction Accuracy

As for the prediction accuracy results, we first show the Root Mean Squared Error (RMSE) of the four methods, since RMSE are one critical factor for evaluating the prediction model, and the calculation is shown as follows.

$$\bullet \text{ } RMSE = \sqrt{\frac{1}{|S|} \sum_{i=1}^{|S|} (z^i - \hat{z}^i)^2}$$

Figure 2. The results of time overhead against the number of English syntax texts



The corresponding results are presented in Fig. 3. In particular, we can see that DTM achieves the smallest value of RMSE. RMSE indicates the error rate between the actual value and the prediction value. For any prediction model, we want to reduce the difference between the prediction result and the actual result, such that the smaller the value of RMSE, the better. Taking this into consideration, DTM performs the best in terms of the prediction model. SVM achieves the second smallest value of RMSE, which is relatively close to that of DTM. Although they have similar RMSE, their training processes differ a lot.

Based on the above value of RMSE, we now present the results of the prediction accuracy. Apparently, we can see that the results in Fig. 4 show the similar trend as in Fig. 3. That is because the prediction accuracy is highly proportional to the performance of RMSE. On one hand, the method with smaller value of RMSE will achieve higher prediction accuracy. For example, DTM has the smallest value of RMSE (about 0.69 on average) and it achieves the highest prediction accuracy (about 95.4%). On the other hand, any two methods with similar RMSE will achieve similar prediction accuracy. For example, the prediction accuracies of DTM and SVM are close, that is, 95.4% and 93.2%. Moreover, TSM and KNM have lower prediction accuracy, that is, 85.8% and 87.1% respectively.

4.2.1 Result Over Different Platforms

Lastly, the proposed method mainly relies on using the random forest and the decision tree. Considering the fact that there are a lot of platforms that have already implemented such methods. We then leverage different platforms that have embedded the random forest and decision tree methods, to evaluate the efficiency of DTM on different platforms. In order to comprehensively show the performance, four platforms are selected, that is, the scikit-learn, the Spark MLlib, the DolphinDB, and the XGBoost over the indicators of training speed, memory accuracy and the prediction accuracy. The corresponding results are shown in Table 3. Apparently, although these platforms use the same method DTM, their efficiencies are totally different. For example, XGBoost only demands 0.48GB memory, while Spark MLlib demands 134.6GB memory. On the contrary, XGBoost needs a long training time, that is, 103.8 minutes, while Spark MLlib only needs 20.1 minutes. Nevertheless, their prediction accuracy are almost the same and this is reasonable, because the algorithm essence of these platform are the same.

Figure 3. The results of RMSE

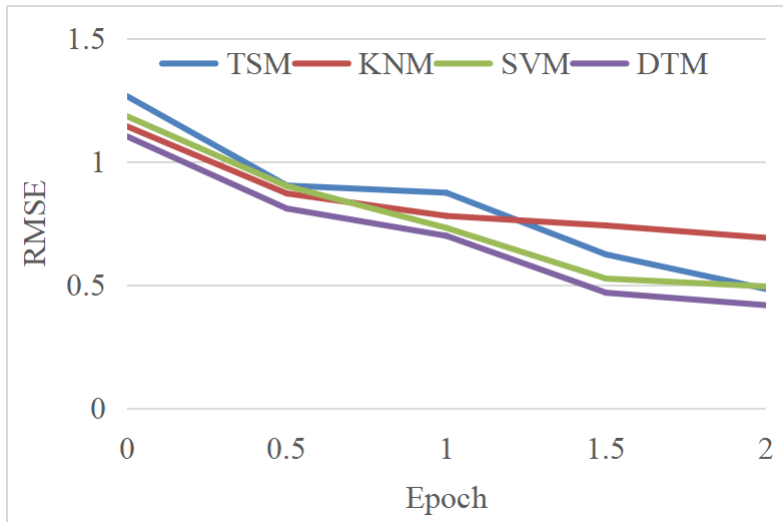
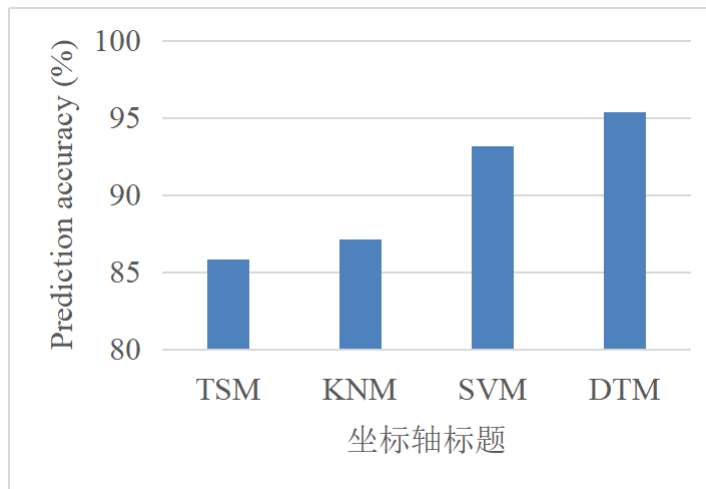


Figure 4. The results of prediction accuracy



5. CONCLUSION

In this work, we study the intelligent English teaching method to further improve the quality of English teaching. Specifically, it consists of several parts. The first part is to implement the data pre-processing, which makes sure that the English syntax texts are in the same format. The second part uses the random forest to establish a training model. The last part relies on using the decision tree to make a prediction about the English text in terms of its grammar or syntax issues. The evaluation results indicate that the proposed method can effectively improve the accuracy of English grammar or syntax recognition and outperforms the other benchmarks greatly.

Table 3. Results of DTM over different platforms

Platform	English syntax text data	Time (min)	Memory (GB)	Accuracy
scikit-learn	100000	78.3	1.1	0.949
Spark MLlib	100000	20.1	134.6	0.956
DolphinDB	100000	56.9	3.3	0.948
XGBoost	100000	103.8	0.48	0.952

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Prediction of Football Match Results Based on Edge Computing and Machine Learning Technology

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ABSTRACT

With the rapid development of artificial intelligence, various machine learning algorithms have been widely used in the task of football match result prediction and have achieved certain results. However, traditional machine learning methods usually upload the results of previous competitions to the cloud server in a centralized manner, which brings problems such as network congestion, server computing pressure, and computing delay. This paper proposes a football match result prediction method based on edge computing and machine learning technology. Specifically, the authors first extract some game data from the results of the previous games to construct the common features and characteristic features, respectively. Then, the feature extraction and classification task are deployed to multiple edge nodes. Finally, the results in all the edge nodes are uploaded to the cloud server and fused to make a decision. Experimental results have demonstrated the effectiveness of the proposed method.

KEYWORDS

Edge Computing, Football Result Prediction, Machine Learning, Sport Match Result Prediction

1. INTRODUCTION

Football is one of the most popular sports. Predicting the results of football matches is interesting to many, from fans to punters. It is also interesting as a research problem, in part due to its difficulty, because the result of a football match is dependent on many factors, such as a team's morale, skills, current score, etc. So even for football experts, it is very hard to predict the exact results of football matches (Owramipur 2013). However, since various types of football matches have outstanding similarities in some respects, in theory, it is possible to find the law from a large number of football matches to find a way to judge the level of victory or defeat (Guan 2021).

With the rapid development of artificial intelligence, machine learning algorithms have been widely used in real life, such as face recognition, stock price prediction, etc. Hence, how to build a football match prediction model based on machine learning algorithms and use scientific methods to solve the prediction problem has become a topic of interest to experts and scholars. Fortunately, researchers have constructed a variety of football match result prediction methods, and have achieved some results. For example, literature (Igiri 2015) used the SVM algorithm to study the factors that affecting the results of the British Championship. Lei used the logistics regression algorithm in machine learning to analyze and process the historical results of football matches, and realized the prediction of the football match results (Lei 2019). Guan also achieved effective result prediction of the Chinese Super League team by suing fuzzy neural network and extreme learning machine (Guan 2021).

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However, traditional machine learning methods usually use centralized computing, which need to upload previous matches results to the cloud server for centralized processing. When there is a lot of data to be processed, the current methods will face new problems, that is, with the increase of match data, the amount of data that needs to be uploaded to the cloud server will increase significantly, which will occupy a large amount of bandwidth and cause network congestion. And if the classification tasks are all completed by the cloud server, the computing resources cannot be dynamically scheduled, which will occupy a large amount of server computing resources. In addition, in order to improve the learning ability of the classification model, it is usually necessary to increase the number of parameters. However, the increase in the amount of model parameters will cause the classification model to occupy too much computing and storage resources, which will further increase the computing pressure of the server.

In order to solve the above problems, this paper proposes a football match result prediction method based on edge computing and machine learning technology. Edge computing is a new computing method that appears to solve the problems of network congestion and cloud center computing pressure faced by cloud computing. Specifically, edge computing realizes the nearby processing of data by deploying edge computing nodes at the end close to the collection device, and these nodes have the capabilities of computing, storage and communication. And this manner can effectively alleviate network congestion, and sharing the computing tasks of the cloud center by edge nodes can effectively reduce the computing pressure of the cloud center (Wang 2020). Inspired by this, this paper splits the football match result prediction task into some sub-tasks that can be executed at multiple edge nodes, so that the task of feature extraction and classification is deployed on the edge computing nodes.

The remainder of this paper is organized as follows: Section 2 gives a summary of previous work on football result prediction. In Section 3, we describe the proposed architecture of football match result prediction method based on edge computing and machine learning technology. The experiments are provided in Section 4. Section 5 is the conclusion.

2. RELATED WORKS AND ANALYSIS

With the increasing popularity of football, the problem of predicting football match results has become a hot topic in the field of football sports, and it is also an important research content in academic circles. As early as 1990, a football result prediction modal was first proposed based on the average goal rate per football match at the International Gaming Conference. With the progress of the times and the development of technology, there are more and more ways to predict the results of football matches (Arabzad 2014). It can be seen from this that with the popularity of football worldwide, the commercialization of football has also entered a fierce struggle. Whether it is a fan or a data analyst who is interested in football matches, their research on predicting the results of football matches has never stopped. As a result, based on strict mathematical statistics, many scientific methods have been emerged. According to different fields of researchers, there are mainly the following prediction methods.

Bayesian model: The probability of both sides winning the match is determined by various uncertain factors such as the strength of both sides, including the number of historical battles and records of the two teams, and offensive and defensive capabilities. These three factors are used to compare the relative strength of any teams that ultimately scores more than the competing team. Other factors related to team strength may also affect the winning rate, but the Bayesian model believes that their influences are very small. The disadvantage of the Bayesian model is that it places too much emphasis on historical records. With the development of the times, it is impossible to predict the outcome of the current match by only relying on past play data (Razali 2017).

Fuzzy comprehensive evaluation: The fuzzy comprehensive evaluation method analyzes the factors that are likely to attract everyone's attention in the football match as an index to predict the football results, these factors include the number of goals, the number of shots, the weather temperature,

the overall state of the team. However, fuzzy comprehensive evaluation requires too many indicator dimensions. The difficulty of using this type of method to predict the football match result is the data acquisition and data cleaning in the early stage (Cao 2011, Rao 2014).

Machine learning/data mining: In today's rapid development of artificial intelligence, how to build a football match prediction model based on machine learning algorithms and use scientific methods to solve the prediction problem has become a topic of interest to experts and scholars. Machine learning algorithms can obtain features from the original data through feature extraction operations, and try to obtain the relevant laws of the data, then use the obtained laws to predict the unknown data set (Sun 2017). Commonly used machine learning algorithms mainly include k-nearest neighbor algorithm (KNN), support vector machine (SVM), logistic regression (LR), decision tree, random forest, etc. (Mlouk 2017). SVM can effectively adjust the contradiction between algorithm complexity and generalization ability, so it has better promotion ability than traditional pattern recognition methods in the field of small sample learning (Yao 2017). However, when dealing with larger data sets, it usually takes longer to train. KNN is an example-based learning method that can generate decision boundaries of any shape without the need to build a model, but its classification overhead is very large, because similarities need to be calculated one by one. In addition, when the value of k is small, it is also very sensitive to noise (Chatzigeorgakidis 2017). In response to the above shortcomings, the researchers have made many improvements, but there are no classification methods that can simultaneously achieve short training time, strong predictive ability, easy rule extraction and strong adaptability. In comparison, logistic regression has better performance. Because the data characteristics in the football match, such as the number of shots, the number of free kicks in the front court, the number of corner kicks, the number of goals, etc., have an obvious non-linear relationship with the result of the match, the neural network method is very suitable for dealing with this non-linear prediction problem. Guan achieved effective result prediction of the Chinese Super League team by using fuzzy neural network and extreme learning machine (Guan 2021). Literature (Sujatha 2018) realized football match result prediction based on BP neural network, and analyzed various factors that affect the match result based on historical data.

However, traditional machine learning models need to upload the match results to cloud server in a centralized manner, and the cloud server performs classification calculations separately, which is likely to cause network congestion and occupy a large amount of computing resources on the server. Fortunately, edge computing can effectively solve this defect.

Based on the enlightenment of the above content, facing the problem of football match result prediction, borrowing the idea of edge computing, this paper established three different prediction models based on KNN, logistic regression and neural network, respectively.

3. ARCHITECTURE OF FOOTBALL MATCH RESULT PREDICTION METHOD BASED ON EDGE COMPUTING AND MACHINE LEARNING TECHNOLOGY

3.1 Problem Formulation

From the cybernetic point of view, the task of creating football result prediction models reduced to find out functional mapping from input features to match results:

$$X = \{x_1, \dots, x_t\} \rightarrow D \in \{d_1, d_2, d_3\}, \quad (1)$$

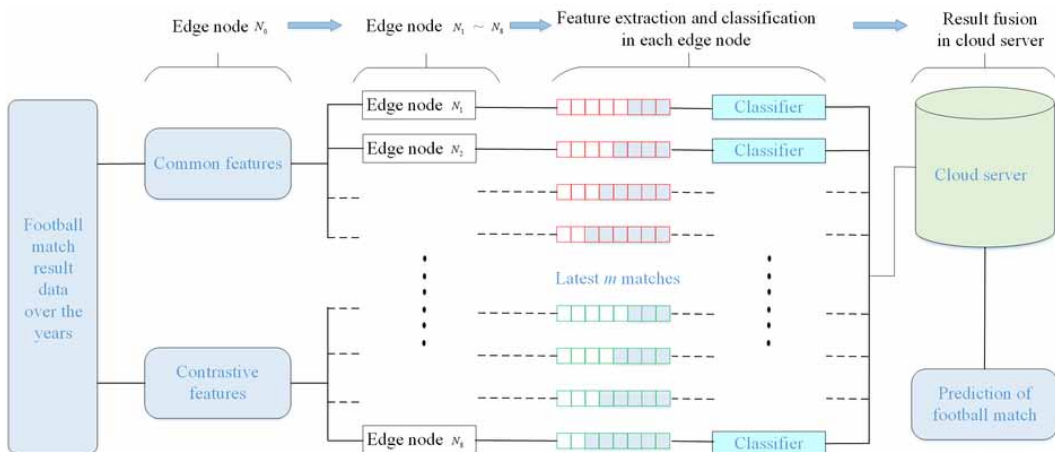
where, X denotes a vector of features (i.e. influence factors), such as number of goals, the number of shots, etc. D denotes the football match result for assessment of one of the terms, and d_1 denotes the host team's win, d_2 denotes draw, d_3 denotes the guest team's win, usually $d_1 = 1$, $d_2 = 0.5$, $d_3 = 0$.

3.2 Architecture of The Proposed Framework

In order to solve the problems of traditional machine learning-based football match result prediction methods that cannot respond in real time, occupy storage and data transmission volume is too large, etc., an architecture of football match result prediction method based on edge computing and machine learning technology is proposed, which is shown in Figure 1.

In Figure 1, the architecture proposed in this paper can be divided into two parts: Edge computing nodes (hereinafter referred to as edge nodes) and cloud server. Among them, edge nodes undertake the main computing task. Specifically, the N_0 node extracts features from historical match results: Including past match results features with all teams (called common features), and match results features with opponent teams (called contrastive features). And transmit these features to edge nodes $N_1 \sim N_m$. On this basis, edge nodes $N_1 \sim N_m$ continue to extract features of different time scales, where $m = 8$. Specifically, edge nodes $N_1 \sim N_4$ are used to extract common features such as the number of points, the number of home goals, the number of away goals, etc., from the last 3, 4, 5, 6 matches, respectively. And edge nodes $N_5 \sim N_8$ are used to extract contrastive features such as the number of goals scored, the number of goals conceded, the number of points, the number of home goals, the number of away goals, etc., from the last 3, 4, 5, 6 matches with a specific opponent, respectively.

Figure 1. The illustration of the architecture for football match result prediction



Our method in this article can be divided into two stages: The training stage and the classification (prediction) stage. In the training stage, the edge nodes $N_1 \sim N_m$ respectively upload the extracted features of different time scale to the cloud server, and receive the model F issued by the cloud server. In the classification stage, the edge nodes input the features into the classification model and output the prediction result, and then upload the result to the cloud server. The features in the classification stage are not uploaded to the cloud server, thereby reducing network transmission and alleviating network congestion. Besides, the computing tasks of edge nodes are independent of each other, and adding nodes can obtain more scale features and improve classification accuracy. In addition, when computing resources are limited, the number of edge nodes can be adjusted to achieve dynamic scheduling of computing resources.

3.3 Football Match Result Prediction Method Based on Edge Computing and KNN

After obtaining the above model, combined with different classifiers, different match result prediction models can be realized. And this subsection uses KNN as the classifier. The basic idea of KNN is: In the feature space, if most of the k nearest (i.e., nearest neighbors in the feature space) samples near a sample belong to a certain category, the sample also belongs to this category.

Assuming there are n teams in a football league, then based on the results of the past few seasons, we can construct $n(n-1)$ samples:

$$\{(X^1, y^1), (X^2, y^1), \dots, (X^{n(n-1)}, y^{n(n-1)})\}, \quad (2)$$

where $X^i = [X_1^i, X_2^i, \dots, X_8^i]$ denotes the multi-time scale features (here $X_1^i, X_2^i, \dots, X_4^i$ are common features, $X_5^i, X_6^i, \dots, X_8^i$ are contrastive features), and $y^i = \{0, 0.5, 1\}$ denotes the corresponding label, here 0 represents loss, 0.5 represents draw, and 1 represents win.

When testing, we first construct the multi-time scale features of the two testing teams:

$$X^{t1} = [X_1^{t1}, X_2^{t1}, \dots, X_8^{t1}] \text{ and } X^{t2} = [X_1^{t2}, X_2^{t2}, \dots, X_8^{t2}]. \quad (3)$$

Then calculating their nearest neighbors and getting their corresponding output o^{t1} and o^{t2} . Finally, the team's label y^{t1} and y^{t2} can be calculated by the following equation:

$$\begin{cases} \text{if } o^{t1} > o^{t2}, & y^{t1} = 1, y^{t2} = 0, \\ \text{if } o^{t1} = o^{t2}, & y^{t1} = y^{t2} = 0.5, \\ \text{if } o^{t1} < o^{t2}, & y^{t1} = 0, y^{t2} = 1. \end{cases} \quad (4)$$

3.4 Football Match Result Prediction Method Based on Edge Computing and Logistic Regression

Logistic regression is a generalization of linear regression, and is mainly used to deal with classification problems. The input of logistic regression is the sample features, and the output is the probability that the sample belongs to the positive class, hence it is a standard binary classification algorithm.

Since there are three categories, we first use the one-against-one strategy to construct a three-classification logistic regression model based on the data samples in Eq. (1). And marking them as models F_1 , F_2 , and F_3 . When testing, we first construct the multi-time scale features of the two testing teams, then substituting them into the multi-class logistic regression model, the corresponding output o^{t1} and o^{t2} can be obtained respectively. Finally, using Eq. (4) we can get their labels.

3.5 Football Match Result Prediction Method Based on Edge Computing and BP Neural Networks

BP neural network is generally composed of three parts: input layer, hidden layer and output layer. And each layer is composed of several neurons. The learning of BP neural network is a supervised learning process. Specifically, the model continuously adjusts the parameters in the neural network by learning to propagate the error backwards while correcting the error, and finally achieve or close to the expected mapping relationship between input and output. The learning of BP neural network can be divided into two stages.

Forward propagation stage: The input signal passes through the input layer, passes through the hidden layer to the output layer, and finally produces an output result. In the process of information being propagated, the weights between connected neurons remain unchanged, and the output of each layer of neurons only affects the input of the next layer of neurons. When the difference between the output information of the final output layer and the supervision signal (error signal) is less than the set threshold, no changes will be made when the learning is completed, otherwise the error signal will be propagated.

Back propagation stage: The error signal generated in the network will propagate backwards from the output layer to the input layer. During the propagation of the error signal, the weights between the neurons in the network will have error signals for feedback adjustment. The weights will gradually stabilize during the adjustment process, and finally the BP neural network will gradually stabilize to obtain the most ideal output.

Different from the previous models, in this model, the output label is the probability that the sample belongs to the positive and negative classes, that is, the probability of the team winning and losing, so the output is a 2-dimensional vector. If the team wins, the corresponding label is set as $[1; 0]$. If the team losses, the corresponding label is set as $[0; 1]$. If the team draws, the corresponding label is set as $[0.5; 0.5]$. Where the first element in this vector represents the probability of the team winning, the second element represents the probability of the team losing, and their sum is 1.

After obtaining the corresponding labels, substituting the training data in Eq. (1) into the model, we can get the trained neural network model F_N . When testing, we first construct the multi-time scale features of the two testing teams, then substituting them into F_N , the corresponding output $o^{t1}=[A_1; B_1]$ and $o^{t2}=[A_2; B_2]$ can be obtained respectively. Finally, the team's label y^{t1} and y^{t2} can be calculated by the following equation:

$$\begin{cases} \text{if } A_1 - A_2 > 0.5, & y^{t1} = [1; 0], y^{t2} = [0; 1], \\ \text{if } A_1 - A_2 < -0.5, & y^{t1} = [0; 1], y^{t2} = [1; 0], \\ \text{if } |A_1 - A_2| \leq 0.5, & y^{t1} = y^{t2} = [0.5; 0.5]. \end{cases} \quad (5)$$

4. Experimental results and analysis

This section includes four parts. The main objective of the first part is to introduce the database. In the second part, we will describe in detail how to construct data features. The football match result prediction experiments are performed in the third part. In the last part, we will conduct ablation experiments to verify the influence of edge nodes on the prediction results. All experiments are carried out on a PC with system configuration Inter(R) Core (TM) 2.60GHz with 8GB RAM.

4.1 Database

In this subsection, we use the Chinese Football Super League for the seasons 2008-2018 as an example to verify the effectiveness of the proposed method. In this league, there are 16 teams in each season. However, in the 2008-2018 seasons, there are only 11 teams that have participated in at least 8 seasons, and in our experiments, we select 10 teams to construct our database. Thus, according to Eq. (1) we can construct totally 90 data samples. In the experiment, 80 samples were selected as the training set, and the remaining 10 samples were used as the testing set.

4.2 Sample Features

In this article, there are two types of features need to extract, namely common features and contrastive features.

The common features are extracted from the results of all matches in the past few seasons, their elements mainly include: the transfer money spent, League rank, total goals scored, total goals conceded, home goals scored or conceded, away goals scored or conceded, team cost, year of match, League points, home points, away points, home advantage (if the match is being played at home for team A, feature value of A will be 1 and correspondingly will be 0 for team B) 14 factors in total.

The contrastive features are extracted from the results of matches against specific opponents in the past few seasons, their elements mainly include: the total goals scored, total goals conceded, home goals scored or conceded, away goals scored or conceded, mutual points, home points, away points, home advantage, total shots, total ball possession 12 factors in total. In addition, in order to reduce the influence of different feature sizes on the experimental results and speed up the convergence speed, the features will be normalized first.

4.3 Experimental Results

In KNN model, the nearest neighbor algorithm is used, i.e., set $k = 1$. For neural network model, the number of hidden layer nodes of the network is set to 32, and the Adam optimizer with learning rate of 0.01 is used. Then, the football match result prediction method based on edge computing and KNN, football match result prediction method based on edge computing and logistic regression, and football match result prediction method based on edge computing and BP neural networks constructed in this paper are used to predict the results of these games, and the final fusion result is obtained by averaging the output. The results obtained are shown in Table 1.

Table 1. Prediction results of different methods

Methods	KNN		Logistic regression		BP neural networks	
	Number	Rate	Number	Rate	Number	Rate
Correct prediction	4	40%	6	60%	8	80%
Wrong prediction	6	60%	4	40%	2	20%

From the prediction results in Table 1, we can observe that among the models constructed in this article, the KNN-based model has the lowest accuracy in predicting the result of the matches, and the logistic regression-based model has a higher prediction accuracy than the KNN-based model. The prediction accuracy of BP neural networks-based model is 80%, which is a higher probability in football match prediction. This means that our proposed BP neural networks-based model is very suitable for predicting the football match result.

4.4 Ablation Experiments

In this subsection, based on the BP neural network model, different numbers of edge nodes are used to verified the influence of edge computing on prediction accuracy. And three experiments are designed: Use common features and contrastive features at the same time, only common features are used, and only contrastive features are used. The experimental results against varying number of edge nodes are shown in Figures 2-4.

From these figures, it can be observed that increasing the number of edge nodes can effectively improve the prediction accuracy, or the use of different time scale features can effectively improve the prediction accuracy. In addition, we also find that when the number of nodes is small, the contribution of contrastive features is greater. As the number of nodes increases, the contribution of common features gradually increases.

Figure 2. The prediction accuracy versus the variation of the number of edge nodes when both common features and contrastive features are used

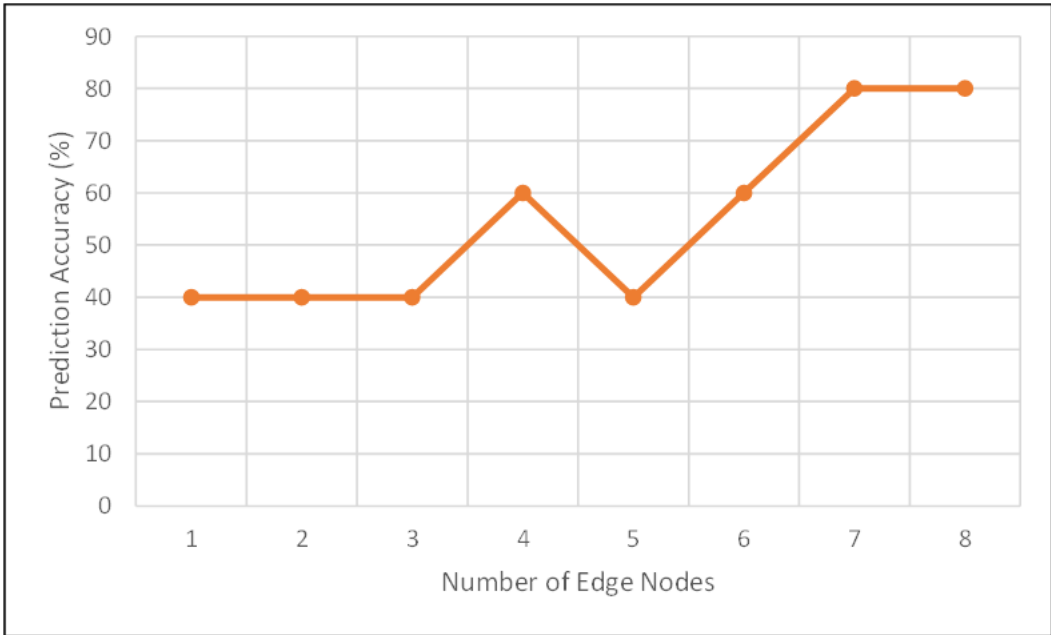
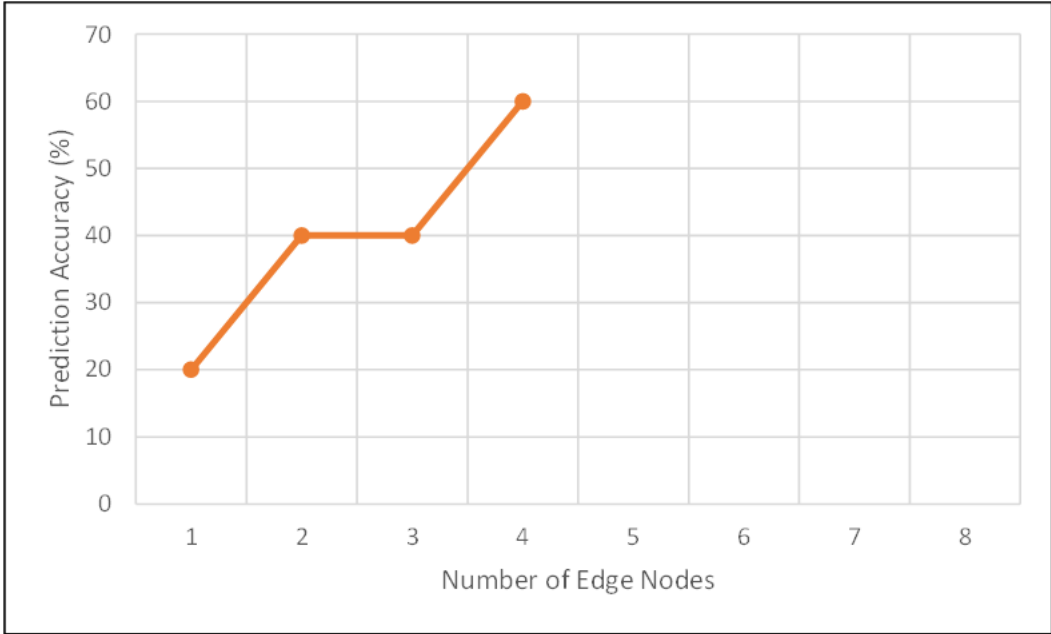
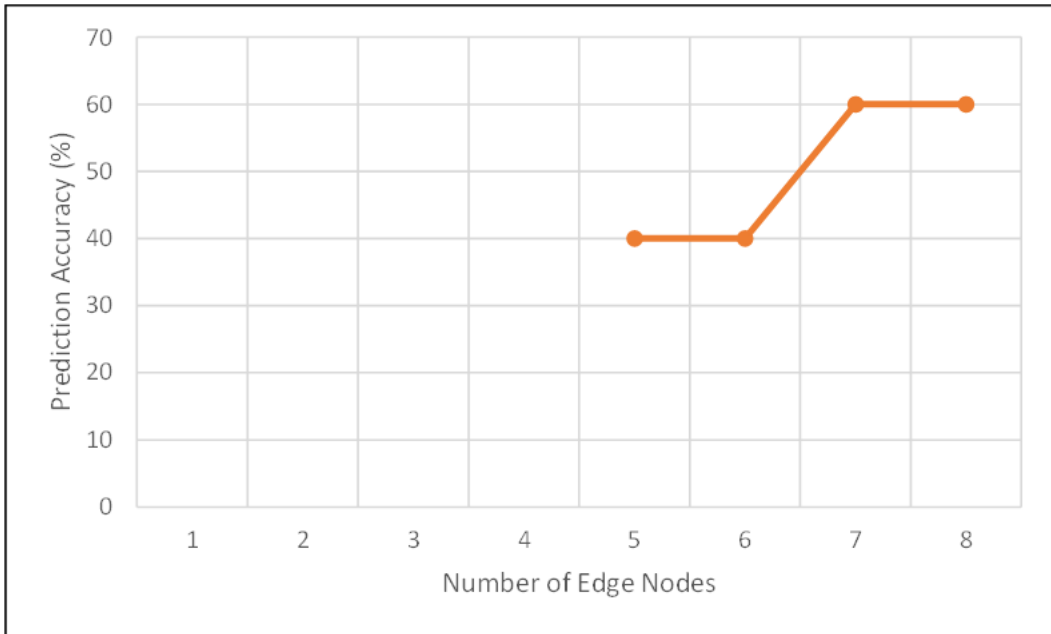


Figure 3. The prediction accuracy versus the variation of the number of edge nodes when only common features are used



5. CONCLUSIONS

Figure 4. The prediction accuracy versus the variation of the number of edge nodes when only contrastive features are used



Aiming at the problem of sports match result prediction, this paper proposes a new football match result prediction method based on edge computing and machine learning technology, which has three different classification models by combining edge computing and three machine learning technology, and realizes effective prediction of football match results. Experimental results on real data sets show that the method proposed in this paper can effectively predict sport match results and provide reference for subsequent research. Future research work includes how to construct a deeper model to obtain more accurate prediction results. In addition, we also interested in selecting more discriminative features.

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Video Sequence Analysis for On-Table Tennis Player Ranking and Analysis

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ABSTRACT

With the rapidly development of the scientific research in the field of sports, big data analytics and information science are used to carry out technical and tactical statistical analyses of competition or training videos. Table tennis is a skill-oriented sport. The technique and tactics in table tennis are the core factors to win the game. With the endlessly emerging innovative playing techniques and tactics, the players have their own competition styles. According to the competition events among athletes, the athletes' competition relationship network is constructed, and the players' rankings are established. The ranking can be used to help table tennis players improve daily training and understand their ability. In this paper, the table tennis players' rankings are established from their competition videos and their prestige scores in the table tennis players' competition relationship network.

KEYWORDS

Competition Relationship Network, Players' Ranking, Video Sequence Analysis

1. INTRODUCTION

With the rapid development of information technology, artificial intelligence (AI) (Dwivedi et al. 2019; Vaishya et al. 2020), Internet of Things (IoT) and cloud computing (Jahantigh et al. 2019; Haji et al. 2020) have been widely used in social development. Relevant scientific research in the field of sports is also developing rapidly (Козина et al. 2017; Matic et al. 2020). The big data analysis has been widely used to carry out technical and tactical statistical analysis for competition videos. As a skill oriented net competition project, the proper skills and tactics are the core of table tennis for the victory. With the development of table tennis sports, various innovative techniques and tactics emerge one after another. Athletes in various countries have their own competition style. Scientific statistical analysis of various techniques and tactics can help athletes win the competition (Bańkosz & Winiarski 2020; Ma 2020).

The study of table tennis techniques and tactics is the main theme of table tennis sports (Wang et al. 2019; Huang et al. 2021). Among them, scientific research scholars mainly use two statistical methods: manual table statistics and computer statistics for technical and tactical analysis. The initial method of technical and tactical statistical analysis is manual table statistics, which is a way for researchers to design index system according to research needs and use tables for statistical analysis with reference to statistical contents. Manual statistics has the advantages, including convenient material acquisition, less investment, large audience and convenient application. However, it also

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has its disadvantages. The statistical speed is slow and the process is cumbersome. The amount of statistical information is huge and needs the cooperation of many people. There are human factors in the statistical results, which is easy to lead to differences in statistical data. The post-processing of statistical data costs lots of time and is difficult to reuse.

Computer statistics (Ceri 2018; Andersson & Kroisandt 2018) is an intelligent processing method for video analysis based on the statistical purpose and statistical demand through machine learning and video analysis. Computer-aided system (Wang & Jiang 2021; Zhang & Liu 2021) has become a development trend of technical and tactical analysis in athletic sports, such as table tennis sports. A well-designed technical and tactical analysis and statistics system can improve statistical efficiency and video utilization, greatly save time, reduce the cost, avoid subjective assumptions, overcome most defects of manual statistics, and provide relevant data for coaches and athletes.

The intelligent technical and tactical statistical analysis of video data combined with machine learning has been widely applied in athletic sports training and competition (Herold et al. 2019; Wenninger et al. 2020). Table tennis statistical index refers to the statistical content designed according to the research purpose to cover the utilization of techniques and tactics of table tennis players in the competition. The difference of statistical needs, statistical means and statistical objects will lead to the difference of statistical objectives. The collection and statistics of table tennis technical and tactical data need experienced researchers or researchers with a deep understanding of table tennis technology and skills. By combining with table tennis technical and tactical analysis theory, the coacher can design statistical indicators to help athletes grasp the correct technical and tactical characteristics.

There are many factors to win table tennis competition. Among them, the control of landing point is an indispensable point of scoring. Between the rounds of each ball, the landing point of the previous ball will have a relevant impact on the threat of the other party's return ball. A good hitting landing point is very important in the table tennis competition.

Statistical index is the embodiment of the concretization of statistical content and the core issue of technical and tactical analysis. It reflects the concept and specific value characteristics of the overall phenomenon. With the help of the analysis of index system, it can effectively reflect the essential characteristics and laws of technical and tactical in the competition to construct statistical index system for table tennis techniques and tactics. The statistical index system should follow the scientific principle, comprehensiveness principle, and measurement principle.

The scientific principle refers to that the design of technical and tactical video statistical indicators should aim at the special characteristics of table tennis, take scientific thought as the guidance, be able to establish a real, accurate, objective and scientific index system, take the statistical purpose and demand as the standard, take the corresponding discrimination standard as the evaluation scale, and take the expert consultation and questionnaire results as the basis with clear content and mutual independence to build a scientific and feasible index system.

The comprehensiveness principle refers to that the statistical index system should fully cover all statistical indicators to make it be comprehensive. According to the different statistical objects and statistical needs, the statistical indicators adopted by researchers are also different. Through consulting the literature, it can be seen that more researchers focus on the statistics of technology and tactics, and adopt the three-stage index or four-stage index evaluation method. The landing point in table tennis competition is not only the key point of scoring, but also a measurable index. Simply analyzing the use of technologies and tactics cannot fully explain the situation of the game.

The measurement principle requires the selected index must be objective and qualitative to avoid the uncertainty of statistical results. The statistical index data is an intuitive embodiment of the situation of athletes in the sports. The index content is the key factor to build a table tennis technical and tactical video statistical system. The meaning of the index needs to be clear. The data should be easy to collect.

The primary indexes are applied to the actual measurement of the sports team, and the index contents are revised according to the modification opinions. Through the questionnaire survey to test

the rationality of the index system, we can screen the failed indexes and improve the contents covered by the index system. The questionnaire contains three criteria level indicators of tactics, technology and landing point. It can more accurately express expert opinions and opinions, reflect the results of the questionnaire survey, achieve the purpose of the questionnaire survey, and is conducive to the accuracy and reliability of the whole questionnaire survey results. This paper considers tactical indicators, technical indicators, landing point indicators and gain and loss indicators by combining with the special characteristics of table tennis and taking the video technical and tactical statistical indicators of table tennis competition as the starting point. Then, the index system is constructed according to technical and tactical video statistics for table tennis competition. The establishment of the index system lays a solid foundation for the construction of table tennis technical and tactical video statistics system to ensure the smooth development of the follow-up statistical work. Then, the results from index system are further used to analyze the ranking of the table tennis players.

Based on the node centrality research of complex network (Saleh et al. 2018; Baravalle et al. 2019), this paper proposes athlete's ability ranking through the analytical results of the training and competition videos of table tennis players to show their potential competition ability. Firstly, we collect all the table tennis training and competition videos. The analysis results are converted as the network relationship. The times between athletes in training or competition are used as the edge weight. Then, the athlete ranking is proposed to quantify the importance of the athletes by using the prestige score of the athletes. Finally, the athletes are ranked through the experimental data, and then compared with the ranking given by coaches or experts to verify the effectiveness of the proposed ranking method.

2. COMPUTER-AIDED SYSTEM TO ASSIST TABLE TENNIS PLAYER TRAINING THROUGH COMPETITION VIDEO AND DATA ANALYSIS

A social system often has complex characteristics. The complexity is generally shown at that the human behavior often follows complex dynamic patterns, such as behavior deletion in the establishment of electronic communication rules. At the same time, the complexity is also reflected in the overall level. For example, when the social system is represented in the mathematical form of graph or network, the node represents the individual, and the edge represents the interactive relationship between individuals in social activities. In most cases, social networks are scale-free from the perspective of topology and global.

In recent years, complex networks have become an important approach to study social system problems and a hot topic in the research of interdisciplinary (Ding et al. 2019; Cherifi et al. 2019). Current social activities release a large amount of data to describe human activities and their interaction. With the development of data science and network science, it has become an important task of artificial intelligence research by using big data technology to analyze the interdependence between massive data, mine the internal laws of data and reflect the relationship of data objects. For the current social system, a new trend is to analyze the macro statistical characteristics of social interaction system composed of a large number of individuals other than the behavior of a single individual. The aim is to find and apply the laws. From this aspect, large-scale data about sports events are processed by complex network, such as football, rugby, baseball, and basketball. In this paper, the complex network is adopted to analyze the table tennis training or competition data.

The table tennis players are used as the network topology nodes and the competition events between players are used as the edge relationship to construct a network topology which is used to describe and mine the big data in the table tennis events.

Obviously, it is a very natural expression in the social system that the competition can be regarded as the basic contact between two players. When a table tennis player i wins the player j in the competition, there is a contiguous relationship between these two players to form a peer-to-peer network topology. The edge weights are adopted to further mine the topological relationship. The edge weight is equal to the number of competition events between two table tennis players.

Then, a non-peer-to-peer network topology is constructed. The constructed network with weights has a high degree of freedom, which can allow the display of table tennis competition relationship networks in various time periods, levels and types. Generally speaking, the network aggregated from a large amount of competition or training data has similar network topology complexity characteristics with the social network. The classical method to measure complex networks is to calculate the probability density distribution of node degree, and the result follows a significant power-law distribution. In the athlete competition relationship network, it means that most athletes only participate in part of the competition or training. A few top athletes will participate in most competitions and beat strong opponents.

This paper aims to rank athletes on the basis of complex network topology. In the network, each athlete has a prestige score to evaluate their ranking in the table tennis sports. Let us assume that the prestige is a variable in the edge weighted network and athletes can improve the ranking by constantly obtaining other people's prestige scores. The process can be solved by establishing the following equation to obtain the system solution:

$$p_i = (1-q) \sum_j p_j \frac{w_{i,j}}{k_j} + \frac{q}{N} + \frac{1-q}{N} \sum_j p_j \delta(k_j) \quad (1)$$

In the Equation (1), i or j represents a node in the complex network, p_i satisfies $\sum_j p_j = 1$, N represents the number of table tennis players, k_j represents the weight of node j . The p_j can be regarded as the prestige score function, which represents the ranking of player j in all table tennis players. The $q \in [0, 1]$ is a control parameter to balance the contribution of each node to the complex network. The term, $(1-q) \sum_j p_j \frac{w_{i,j}}{k_j}$ represents the score proportion of node i during prestige diffusion process. It means that each node redistributes its personal credit to its neighbor nodes depends on the edge weight between two nodes. The $\frac{q}{N}$ reflects that the prestige of table tennis players is evenly distributed to all nodes, and then each player in the network receives a fixed and equal number of credit prestige score. The term, $\frac{1-q}{N} \sum_j p_j \delta(k_j)$, is equal to 0 when $q = 1$.

For a general network topology, the Equation (1) is difficult to be obtained by numerical calculation because only the stable value of Equation (1) is the result of infinite recursive operations. This paper initializes $p_i = \frac{1}{N}$ to iteratively calculate Equation (1) until it converges to the stable value with a priori fixed accuracy a .

In the athlete competition relationship network, the simplest network topology is the result extracted from the data of a simple team training. For this kind of network, the solution of Equation (1) can be obtained by analyzing the related data. In individual championships, the competition relationship network is organized as a hierarchical structure, such as a binary tree. Let r represent the depth of table tennis players, then $N = 2^r$. The Equation (1) can be simplified as a function about the number of competitions r with promoted athletes. Then, the Equation (1) is rewritten as follows:

$$p_r = p_0 + (1-q) \sum_{u=1}^r (p_u - 1) \quad (2)$$

In the Equation (2), $p_0 = \frac{1-q}{N}p_1 + \frac{q}{N}$ and $0 \leq t \leq 1$. The score function p_r is the sum of two parts. The first term, p_0 , represents the average contribution of all qualified players. The second term, $(1-q) \sum_{u=1}^r (p_u - 1)$, represents the score of players' promotion and winning competition. The following iterative relation holds:

$$p_r = (2-q)p_{r-1} = \dots = (2-q)^r p_0 \quad (3)$$

From the Equation (3), it can be found that the Equation (3) relies on a constant value and satisfies the following normalization condition:

$$\sum_{r=0}^l n_r p_r = 1 \quad (4)$$

In the Equation (4), n_r represents the number of athletes who win r competitions. For individual championship, $n_r = 2^{l-r-1}$ holds ($0 < r < l$). Through $\sum_{r=0}^u x^r = \frac{1-x^{u+1}}{1-x}$, the initialized value of p_0 can be obtained by the following equation:

$$p_0 = \frac{q}{2^l + (2-q)^l (q-1)} \quad (5)$$

By combining the Equation (3) and (5), we can obtain the following equation:

$$p_r = \frac{q(2-q)^r}{2^l + (2-q)^l (q-1)} \quad (6)$$

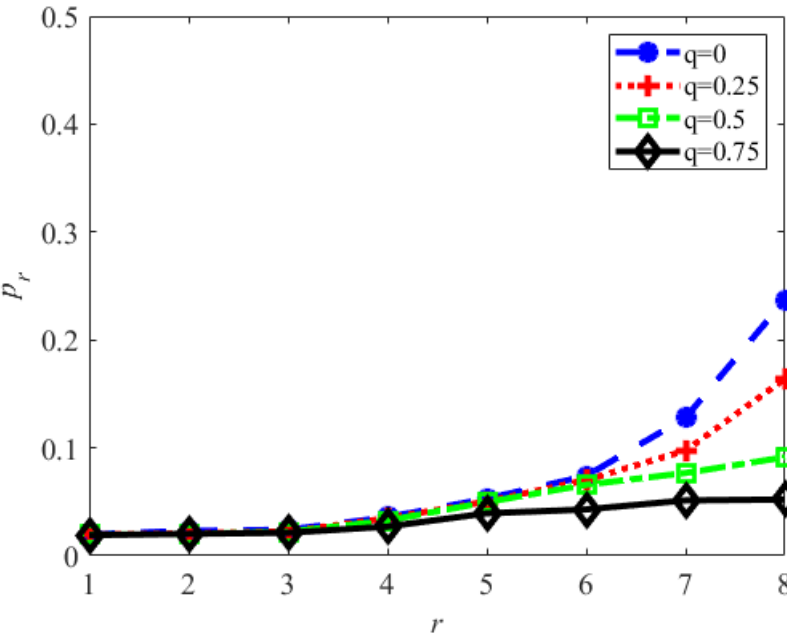
It should be noted that if $q = 1$, the Equation (6) can be simplified to $p_r = 2^{-l}$, which shows that prestige is homogeneous and evenly distributed in the network without diffusion. On the contrary, if $q = 0$, the Equation (6) is transformed as the following equation:

$$p_r = \frac{2^r}{2^l (l+2)} \quad (7)$$

3. EXPERIMENTS AND SIMULATIONS

In this section, we collect the daily training and competition videos of table tennis players from a university. We collect 50 college students' daily training and competition data. The nodes in the complex network depict the information of the table tennis players in daily training and competition videos. The following figure illustrates the relationship between p_r and r .

Figure 1. The distribution of p_r with different parameter q



In general, a low parameter q makes the players in last round of competition have two orders of magnitude higher scores than the opponent in the first round. The best player should be in the form of exponential function of the number round in competition, l . In our experiment, the parameter q is set as 0.25. Table 1 shows the analysis of the athlete competition relationship network constructed by the table tennis training and competition data, including the prestige ranking of the top 10 athletes. In order to protect the privacy of the college student players, the Table 1 only lists the surname of the players.

Then, the results in Table 1 are compared with the evaluation of coaches, which shows the prestige scoring method based on complex network proposed in this paper can effectively rank athletes in

Table 1. The top 10 prestige ranking of table tennis players in universities

Order	Name	Uni.
1	Zhang **	PKU
2	Liu **	BJUT
3	Wang **	RUC
4	Qian **	BFSU
5	Chen **	THU
6	Cheng **	BNU
7	Xu **	CUC
8	Zheng **	USTB
9	Yang **	CUFE
10	Hu **	BJRU

Figure 2. The prestige ranking of top 10 table tennis players in universities

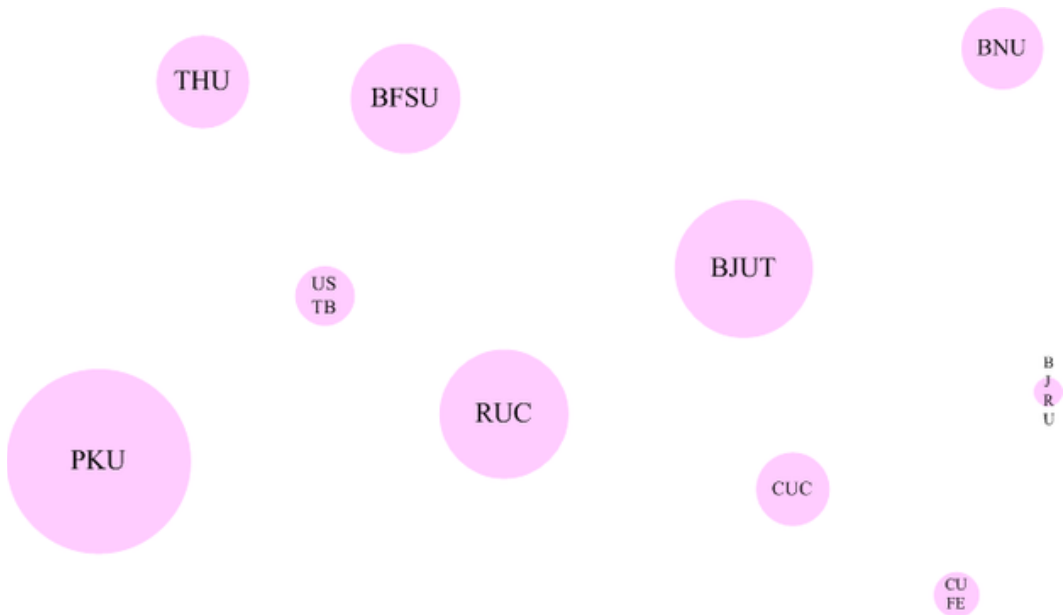


table tennis training and competition. Through the above analysis, it can be found that this algorithm has better advantages in showing the current competitive state of table tennis players. The following figure shows the prestige ranking of top 10 players. Each circle represents a player. The size of the circle reflects the associated ranking.

4. CONCLUSION

In this paper, we propose a framework to evaluate the prestige ranking of table tennis players and assist daily table tennis teaching and training in university. First, we collect the daily training and competition videos of college students during table tennis teaching and training. Second, according to the results of competitions in daily teaching and training. Third, the table tennis players are represented as the competition data and used as the nodes in complex network. Lastly, the prestige ranks of table tennis players are constructed by using complex network model. The experimental results on real university table tennis teaching and training data show that the proposed framework can accurately describe the prestige ranks of the table tennis players, which is helpful to assist daily table tennis teaching in university.

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Aesthetic Evaluation of Interior Design Based on Visual Features

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ABSTRACT

In the modern context, interior design has inevitably become a part of social culture. All kinds of modeling, decoration, and furnishings in modern interior space show people's pursuit and desire for a better life. These different styles of modern interior design rely on science and technology and utilize culture and art as the connotation. Its development often reflects the cultural spirit of a nation. The aesthetic evaluation plays an important role in the modern interior design. With development of derivative digital devices, a large number of digital images have emerged. The rapid development of computer vision and artificial intelligence makes aesthetic evaluation for interior design automatic. This paper implements an intelligent aesthetic evaluation of interior design framework to help people choose the appropriate and effective interior design from collected images or mobile digital devices.

KEYWORDS

Aesthetic Evaluation, Interior Design, Machine Learning, Mobile Digital Device

1. INTRODUCTION

With the development of human society, the development of interior design is also step by step and serves people's life (Grieze & Miķelsone 2021, Hasti & Kusnia 2019). People's life and interior design promote each other with the development of society (Chen & Wang 2020, Afifi 2020). The great difference between design and art is that design pays more attention to practical functions. Different design products have different use functions, and the requirements of different interior design are also different. Because the applicable population and spatial positioning of indoor space are different, the design methods will also be different.

The interior design evaluation refers to the comparison and evaluation of the interior design (Wang & Hsiao 2018, Valiyev 2020). It should be noted that interior design evaluation belongs to a neutral attitude, which is only a scientific evaluation of interior design works. The interior design is the product of people's practical life (Tautkute 2019). In the process of interior design development, there will be cases that do not conform to the people-oriented concept, and the designed products are incompatible with the market, which is also the reason for the emergence of interior design evaluation. In order to maintain the competitiveness of the product itself, the designer must push through the old and bring forth the new. This process also needs design evaluation. In interior design, the designer himself is a design evaluator. He has been in a state of repeated modification for his own design, in which the designer plays an important role in the design process.

The interior design is a relatively large design scope, which contains many elements, such as building decoration materials (Cao 2021, Zeng & Jiang 2018), indoor furniture (Viyanon 2017, Sarkar & Bardhan 2019), and decoration style (Tong 2019, Jinkun 2017). The evaluation methods are also

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different in these different interior design elements. The interior design evaluation is used to compare and comment on design works. For a design work, the evaluation is related to its factors and consumer requirements. For the decoration materials required for interior designer, people pay more attention to its environmental protection on the premise of paying attention to its aesthetics, which is also the difference between it and other design elements. For the evaluation of decoration materials, we should take green environmental protection as the premise and focus on its functionality since the existence value of decoration materials lies in its functionality to support and form the whole indoor space.

For the evaluation method of interior design, we should not only take care the functionality and aesthetics of the furniture itself, but also put the furniture in the corresponding indoor space and watch its overall effect. Furniture has its own design style. The corresponding location in indoor space depends not only on its own style, but also on whether the design style of furniture in the whole interior design is appropriate, which is also the particularity of furniture design evaluation.

For everyone's favorite design elements, the favorite style and suitable style are different. For instance, children need bright and lively colors to suit their childlike innocence. The elder's house needs to be designed with ease, comfort and generosity, which is not suitable for a large number of complex decorations. Each design style needs special evaluation method. For the interior design evaluation, we should first start from people, adopt the people-oriented concept, and carry out scientific evaluation and theoretical analysis. However, as the object of human being is the evaluator, it is inevitable that subjective factors will appear in the interior design evaluation process, which requires our evaluators to make an objective and fair evaluation and give a rational answer to the public and society. In this paper, we adopts image aesthetics to evaluate interior design.

Image aesthetics (Ren 2017, Chen 2020) is an aesthetic way based on image and guided by art. As an important bridge for people to obtain and exchange information, the image is very specific, intuitive, easy to understand, and can greatly enrich human life and work. With the rise of computer and artificial intelligence, more and more electronic products emerge in endlessly, and a large number of digital images are emerging. It is become necessary to automatically divide the images into aesthetic grades and assist human to complete the image processing. Automatic image aesthetics evaluation (Li 2020, Peng 2021) is a research problem with great significance and wide application prospects. The potential application fields involve visual experience, such as image retrieval (Wang 2017), image editing and design (Wang 2019), human computer interaction (Dou 2019). It can be applied to many fields, especially for human-computer interaction. The image aesthetics plays an important role in all aspects related to image. Mathematical images can be applied to different fields of life and science and technology through different ways of processing.

2. INTERIOR DESIGN EVALUATION THROUGH AUTOMATIC IMAGE AESTHETICS

Image aesthetics is an interdisciplinary comprehensive research topic in many fields. At the same time, it is also closely connected with the booming artificial intelligence. It has a very broad application field and development prospect. It is very meaningful to enable the computer to simulate human aesthetic perception, realize the evaluation of image aesthetics, and replace human work in dealing with massive redundancy.

With the rise of computable image aesthetic evaluation, many researchers study different methods to solve the problem of image quality evaluation. It is an urgent problem to represent the image. From a computable point of view, image aesthetic evaluation can be regarded as a process of obtaining aesthetic prediction from an image. The process can be regarded as two parts. One is feature representation. The other is model learning. For the aesthetic research of image, most of the existing literature mainly focuses on the extraction of image features, that is, to extract image features by simulating the composition mode of image or the theory of psychology and perception, so as to realize the aesthetic evaluation of image.

By comparing the research methods in different stages of image aesthetics evaluation, it can be seen that the manual feature method has more feature redundancy information and its implementation

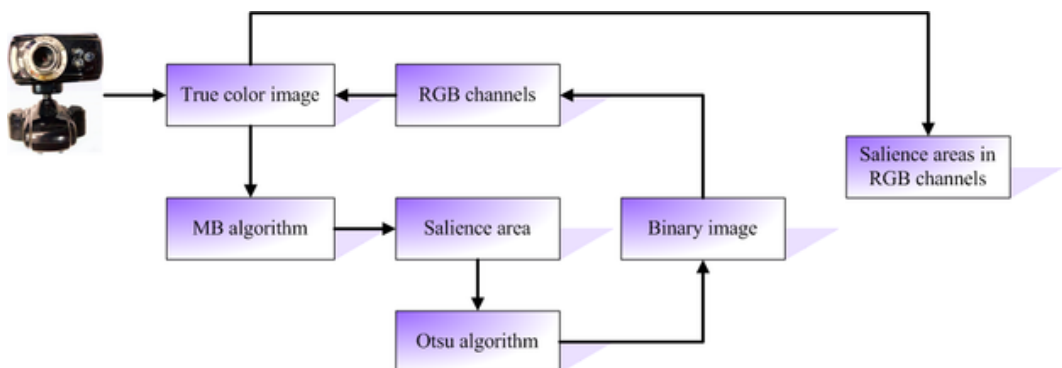
is complex. At the same time, some aesthetic related attributes are difficult to define an objective evaluation standard. Many photographic or psychological rules are descriptive, and the extracted features are only close to these rules. The general features commonly used in image classification and retrieval are applied to aesthetic evaluation directly. Although the implementation effect is better than the underlying manual features, it does not provide a better aesthetic feature descriptor for the problems related to image aesthetics. Although the features learnt by deep neural network (Fu 2018) achieve state-of-the-art results, the deep learning costs many resources to learn the feature representation model which is more complex than previous methods. Additionally, the deep learning needs massive data as training set and strong hardware equipment to support model training.

The image aesthetics evaluation considers both subjective and objective factors. It refers to people's educational background, aesthetic experience, cultural region, personal hobbies and so on. Because the categories of natural scene images are complex and rich, natural images of the same type of scene have aesthetic differences. The images from different scene types are vulnerable to semantic information, which makes the image aesthetic evaluation method vulnerable to human strong subjective factors and different forms of objective factors. Therefore, there is no unified, definite and objective evaluation standard for the evaluation of image aesthetics, which also makes researchers study image aesthetic evaluation from different perspectives.

The visual attention mechanism (Jian 2019) is an important characteristic of the human visual system. Based on this characteristic, many researchers have deeply studied the image saliency region detection algorithm which has been widely used in many fields. The existing research about saliency region detection can be summarized into the following categories: contrast characteristics based methods, spectrum characteristics based methods and information entropy theory based methods.

The pixels at the image boundary are used as the background image. By modeling the background image and using the minimum obstacle distance transformation method, the salient image can be extracted accurately and efficiently. The maximum variance threshold segmentation method (Wang 2018), can automatically determine the best threshold. Through the determined threshold, the image is divided into foreground image and background image, so as to realize the effective segmentation of different natural scene images. The salient region extraction structure is illustrated in Figure 1.

Figure 1. The flowchart salient region extraction



Firstly, it extracts the image salient region, and then the visual salient image is binarized based on Otsu segmentation method (the maximum variance threshold segmentation method) as the mask of the image foreground region. In order to further obtain an accurate foreground region image, firstly, the pixel of the mask plate of the foreground region is flipped as the background region image mask

plate to obtain the background region image. The foreground region image is obtained by subtracting the original image from the corresponding background region image.

For limited perceptual information, human visual attention mechanism has a certain selectivity in information processing. Through perceptual convergence processing of non-stimulus information, we can centrally perceive and process important stimulus information to achieve efficient processing of limited information. This paper uses the principle that the human eye is sensitive to the fluctuation part of the image, and obtains the prior information and fluctuation information in the gray image by using the frequency domain transformation. The procedure is summarized as follows:

Ø The frequency domain amplitude and phase angle of the image are obtained through the following equation.

$$\begin{cases} m(\mu, \gamma) = \text{abs}\left(F\left(f(x, y)\right)(\mu, \gamma)\right) \\ a(\mu, \gamma) = \text{angle}\left(F\left\{f(x, y)\right\}(\mu, \gamma)\right) \end{cases} \quad (1)$$

Ø The logarithmic spectrum of the image is computed by the following equation.

$$L(\mu, \gamma) = \log\left(m(x, y)\right) \quad (2)$$

Ø The spectral residual information of the image is obtained according to the Equation (3).

$$r(\mu, \gamma) = L(\mu, \gamma) - h_n(\mu, \gamma) * L(\mu, \gamma) \quad (3)$$

Ø The image is transformed into spatial domain based on the Equation (4) to obtain the visual saliency map.

$$sr(x, y) = g(x, y) * \left\| F^{-1} \left\{ e^{r+ja} \right\} (x, y) \right\|^2 \quad (4)$$

Here, $f(x, y)$ represents the original pixels of image, F represents the Fourier transform of the image, F^{-1} represents the inverse Fourier transform of the image, $\text{abs}(\cdot)$ represents the complex amplitude in the frequency domain, $\text{angle}(\cdot)$ represents the complex phase angle in the frequency domain, $h_n(\mu, \gamma)$ represents the mean filter with the size $n \times n$, $g(x, y)$ represents the Gaussian filter, and $*$ represents the convolution operation.

The above algorithm extracts dynamic information based on the spectral characteristics of gray image, ignores the important role of color information in salient image extraction. The HSV color space model consists of chroma, saturation and brightness components. The brightness component is obtained by calculating the maximum component in RGB three channels. It is not related to color information. At the same time, human visual perception is closely related to the hue and saturation color components. The saliency images of the above three color components respectively describe different details in the image. This paper superimposes the HSV three channel images to present the details of the saliency map from the perspectives of chromaticity, saturation and brightness. The mean filter size is set as 3, while the Gaussian filter size is set as 10.

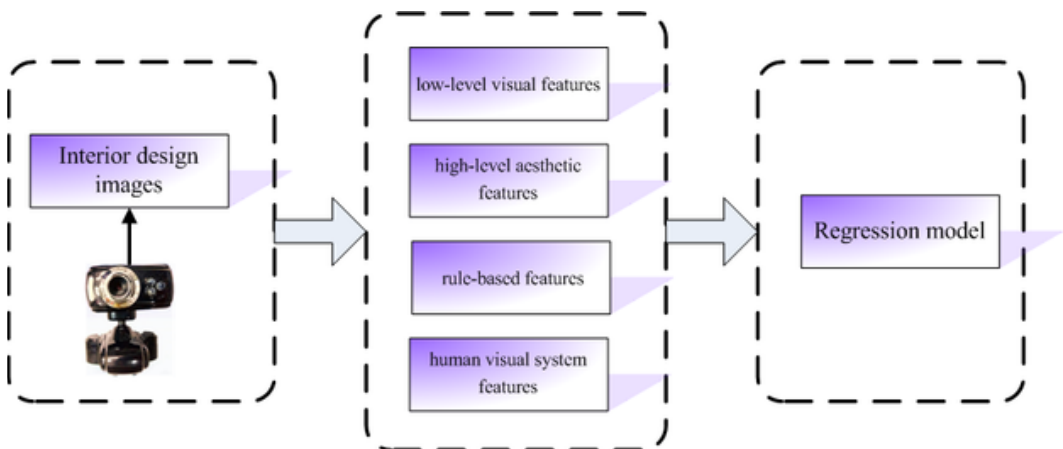
The regression evaluation method of image aesthetics is to score the aesthetics of an image. Compared with binary discrete evaluation method, regression evaluation method is a continuous

evaluation method. Through the aesthetic evaluation of the image, the aesthetic grade of the image is further refined, which poses a further challenge to human simulation and learning by computer. It is an important issue of regression evaluation method to make the aesthetic evaluation of images by computer be close to the evaluation level of human beings and realize the good aesthetic prediction of natural images of different scene types.

Due to the complex influencing factors of image aesthetics, the traditional research methods of manual feature extraction are mainly feature extraction based on single aesthetic perspectives. The research methods based on single features or general features are often difficult to fully capture the different visual aesthetics of images. In order to solve this issue, this paper extracts aesthetic features from the perspective of multi factors by combining the low-level visual features, high-level aesthetic features, rule-based features and human visual system features to describe the image aesthetics features. Natural images of different scene types have their unique aesthetic feeling and the same characteristics have different aesthetic prediction performance for images of different scene types. This paper combines the high-level aesthetic features of different scene types based on the high-level aesthetic features that involve the most aesthetic factors.

The effective and stable multi perspective characteristics and the regressor are two key points in the interior design image aesthetic regression evaluation. This paper uses the combination of optimal feature combination and optimal regressors to predict the aesthetic degree of interior design images. The flowchart is illustrated in the Figure 2.

Figure 2. The architecture of interior design evaluation through digital images



In the Figure 2, the regression evaluation structure consists four parts: input data, feature extraction and selection, regressor selection and optimization, and aesthetics prediction. In the feature extraction and selection part, it considers low-level visual features, rule based features, human visual system features, and high-level aesthetic features. In the regressor selection and optimization, it learns a regressor on a training set and tunes the parameters of the regressor to obtain the best performance.

The low-level visual features include color feature, texture feature and shape feature. The most effective way to attract the observer's attention is the eye-catching and bright color and strong contrast in the image. The color information of the image is highly related to the aesthetic feeling of the image. This paper combines feature map and color moment as the low-level color feature.

When the image contain rich color texture information, the single color information cannot fully describe the aesthetic feeling of the image. At the same time, the aesthetic prediction is disturbed

by various texture information, which results in a large difference between the prediction and the human evaluation. The image texture information plays an important role in studying the image with rich detail information.

Texture can show the roughness and fineness of different regions in the image, and show the overall local detail information of the image. The detail information of the image is usually obtained through multi-scale image. This paper extracts the texture features by combining multi-scale and multi angle detail information.

In general, there is less texture information in the single subject background or blurred background image. The subject target of the image has become the main information affecting the aesthetic feeling of the image. The research on the scene of the image target area plays an important role in describing the scene information of the image. Shape information can directly reflect the scene information and subject contour information of the image, which plays an important role in capturing the spatial composition information of the image and transmitting the intuitive semantic information to the observer.

The color, texture and shape features in the underlying visual features are described from the image color information, rich texture information and target scene information. The rule-based features are different from the underlying visual feature. It describes the aesthetic degree from the aesthetic rules of the image by studying the aesthetic composition mode of the image. This paper extracts two different rule-based features, namely pattern feature and definition feature.

The low-level visual features describe the intuitive visual information in the image. The rule-based features differ from aesthetic modes of the image. The human visual system features are formed based on the human perception system to simulate human visual system. The high-level aesthetic features are extracted different aesthetic factors, including edge distribution feature, depth feature, energy feature, and regional feature.

The edge distribution information of the image reflects the overall complexity of the image. High-quality images often highlight the edge of the main area, so that the observer can intuitively understand the image information.

Professional photographers usually use the method of lowering the depth of field to blur the background area to obtain a high-definition target area, which makes the interest area of the image and the subject target more attractive. This paper adopts the HSV color components. Each component is processed by three-layer wavelet transform to obtain the wavelet coefficient images. The image is divided as 16 blocks to extract the high-frequency information of the central area and surrounding area of different color space, which is written as follows:

$$\begin{cases} w_{3,i}(x,y) = \{w_{3,i}^{lh}, w_{3,i}^{hl}, w_{3,i}^{hh}\} & (i \in h, s, v) \\ LD_i = \frac{\sum w_3(x,y)}{\sum_{i=1}^{16} \sum_{i=1}^{16} w_3(x,y)} & (i \in h, s, v) \end{cases} \quad (5)$$

In the Equation (5), $w_3(x,y)$ represents the high frequency part of wavelet transform coefficients.

Image energy describes the richness and detail information of the image from the perspective of statistics. It can provide a lot of detail information of the aesthetics of the image. The wavelet coefficients (Huang 2017) of the image can reflect the different frequency band information of the image from different transformation levels. Low frequency information contains the main information of the image, the information between high frequency and low frequency, the vertical information and horizontal information of the image. The high frequency information contains the detailed information of different directions of the image. This paper combines the low-frequency image with the high-frequency image from the perspective of main information and rich detail information of the

image to capture the energy information of the image. Based on the HSV three-channel image, the coefficient image after three-layer wavelet transform is obtained. The energy feature of each color component is extracted by the following Equation (6).

$$\begin{cases} w_{3,i}(x, y) = \{w_{3,i}^{lh}, w_{3,i}^{hl}, w_{3,i}^{hh}\} & (i \in h, s, v) \\ LD_i = \frac{\sum w_3(x, y)}{\sum_{i=1}^{16} \sum_{i=1}^{16} w_3(x, y)} & (i \in h, s, v) \end{cases} \quad (5)$$

In the Equation (5), $w_i^{hh}(x, y)$, $w_i^{hl}(x, y)$, and $w_i^{lh}(x, y)$ represent detailed sub-image of the i^{th} layer along diagonal, horizontal, and vertical direction, respectively.

The key areas in the image focus on the key information of the image, and often can intuitively express the image connotation. The observer can quickly understand the information transmitted by the photographer. At the same time, the observer is often attracted by the key areas in the image. How to obtain the key area of the image plays an important role in studying the key information of the image. Generally, the research on key areas often pays attention to the locality of the image, which is easy to make the observer fall into the local image to ignore the image related area information. For the extraction of regional features, this paper extracts regional features from the perspective of local image and global image.

For the regression evaluation method of image aesthetics, the extracted features are used to learn regression model. In this paper, we adopt support vector regression (SVR) (Drucker 1997, Zhu 2017) as the regression algorithm. Let $X \times Y = \{(x_1, y_1), \dots, (x_n, y_l)\}$ represent training set. Here, $x_i \in \mathbb{R}^n$, $y_i \in \mathbb{R}$. The aim of SVR is to find a decision function $f(x) = w^T \phi(x) + b$ by minimizing the following loss function.

$$L_\varepsilon = \begin{cases} 0, & |f(x) - y| \leq \varepsilon \\ |f(x) - y|, & otherwise \end{cases} \quad (7)$$

The objective Equation (7) is implemented by the following optimal programming

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \\ s.t. \quad & \langle w, \varphi(x_i) \rangle - y_i \leq \varepsilon + \xi_i \\ & y_i - \langle w, \varphi(x_i) \rangle \leq \varepsilon + \xi_i^* \\ & \xi_i \geq 0, \xi_i^* \geq 0, i = 1, \dots, l. \end{aligned} \quad (8)$$

The optimal programming is convex. The solution can be obtained by solving the associated dual optimal programming which is written as follows:

$$\begin{aligned}
\min_{\alpha, \alpha^*} \quad & \frac{1}{2} \sum_{i,j=1}^l (\alpha_i - \alpha_i^*) K(x_i, x_j) (\alpha_j - \alpha_j^*) + \varepsilon \sum_{i=1}^l (\alpha_i + \alpha_i^*) - \sum_{i=1}^l y_i (\alpha_i - \alpha_i^*) \\
s.t. \quad & \sum_{i=1}^l (\alpha_i - \alpha_i^*) = 0 \\
& 0 \leq \alpha_i, \alpha_i^* \leq C, i = 1, \dots, l.
\end{aligned} \tag{9}$$

In the Equation (9), $K(x_i, x_j) = \langle \varphi(x_i), \varphi(x_j) \rangle$ in the kernel form of sample x_i and x_j in reproducing kernel Hilbert space (RKHS). The w and decision function $f(x)$ are obtained by the following equations.

$$w = \sum_{i=1}^l (\alpha_i - \alpha_i^*) \varphi(x_i) \tag{10}$$

$$f(x) = \sum_{i=1}^l (\alpha_i - \alpha_i^*) K(x_i, x) + b \tag{11}$$

3. EXPERIMENTS AND SIMULATIONS

In this section, we collect the interior design images from designers. The collected images come from 6 types of interior design. Each image is scored by 20 experts from interior design. The score ranges from 0 to 100. The final score is the average of twenty experts. The number of images from each interior design style are 2,000. The interior design images are divided into two parts equally. One part is used as training set to learn regression model. The other is used as test set to evaluate our interior design evaluation scheme. In our scheme, the interior design images are represented as the combination of low-level visual features, high-level aesthetic features, rule-based features and human visual system features.

The prediction performance of regression model is usually measured by error function. For the evaluation of interior design image aesthetics, this paper adopts root mean square error, average absolute error, correlation coefficient, and Pearson correlation coefficient.

The root mean square error (RMSE) describes the degree of divorce between the real value and the predicted value. The smaller the value is, the higher the prediction accuracy and the better performance the model has. The definition of root mean square error is shown in the following equation.

$$RMSE = \sqrt{\sum_{i=1}^l (y_i - \hat{y}_i)^2} \tag{12}$$

In the Equation (12), n represents the number of samples, y_i is the ground truth of sample x_i , and $\hat{y}_i$ is the predicted value of sample x_i .

The average absolute error (MAE) describes the change degree of the data. The smaller the value is, the more it can reflect the actual situation of the prediction error and the better the performance of the prediction model is. The definition of average absolute error is shown in the following equation.

$$MAZ = \frac{1}{n} \sum_{i=1}^l |y_i - \hat{y}_i| \quad (13)$$

In the Equation (13), n represents the number of samples, y_i is the ground truth of sample x_i , and $\hat{y}_i$ is the predicted value of sample x_i .

The Pearson correlation coefficient (PCC) is a measure of the linear correlation degree between two variables. It represents the similarity between the model and human evaluation. The greater value is, the stronger the correlation between variables and the better the prediction performance of the model is. The definition of Pearson correlation coefficient is shown in the following equation.

$$PCC = \frac{\text{cov}(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}} = \frac{E(y\hat{y}) - E(y)E(\hat{y})}{\sqrt{E(y^2) - E^2(y)}\sqrt{E(\hat{y}^2) - E^2(\hat{y})}} \quad (14)$$

In the Equation (14), n represents the number of samples, y_i is the ground truth of sample x_i , and $\hat{y}_i$ is the predicted value of sample x_i .

The experimental results of root mean square error (RMSE), average absolute error (MAE), and Pearson correlation coefficient (PCC) are reported in Table 1, Table 2, and Table 3, respectively.

From Table 1, the root mean square error (RMSE) of combined features archives 0.4203, 0.3526, 0.3926, 0.3515, 0.3788, and 0.4012 for style 1 to style 6, respectively, while the average RMSE reaches 0.3828. The RMSE of combined features is lower than that of only low-level visual features, high-level aesthetic features, rule-based features, or human visual system features.

Table 1. The root mean square error (RMSE) comparison of different features in interior design

	low-level visual features	high-level aesthetic features	rule-based features	human visual system features	combined features (ours)
Style 1	0.4839	0.4972	0.5013	0.4683	0.4203
Style 2	0.3886	0.4953	0.4419	0.4490	0.3526
Style 3	0.4552	0.4579	0.4977	0.4108	0.3926
Style 4	0.4752	0.4886	0.4478	0.3958	0.3515
Style 5	0.4810	0.4821	0.4918	0.3753	0.3788
Style 6	0.4465	0.4037	0.4274	0.4417	0.4012
Average	0.4551	0.4708	0.4680	0.4235	0.3828

From Table 2, the average absolute error (MAE) of combined features archives 0.2561, 0.2123, 0.2115, 0.2502, 0.2181, and 0.2289 for style 1 to style 6, respectively, while the average RMSE reaches 0.2295. The MAE of combined features is lower than that of only low-level visual features, high-level aesthetic features, rule-based features, or human visual system features.

From Table 3, the Pearson correlation coefficient (PCC) of combined features archives 0.3459, 0.3521, 0.3322, 0.3443, 0.3345, and 0.3514 for style 1 to style 6, respectively, while the average RMSE reaches 0.3434. The PCC of combined features is lower than that of only low-level visual features, high-level aesthetic features, rule-based features, or human visual system features. From

Table 2. The average absolute error (MAE) comparison of different features in interior design

	low-level visual features	high-level aesthetic features	rule-based features	human visual system features	combined features (ours)
Style 1	0.2712	0.2754	0.2804	0.2674	0.2561
Style 2	0.2540	0.2571	0.2293	0.2385	0.2123
Style 3	0.2258	0.2496	0.2451	0.2410	0.2115
Style 4	0.2511	0.2649	0.2595	0.2353	0.2502
Style 5	0.2605	0.2465	0.2566	0.2312	0.2181
Style 6	0.2522	0.2578	0.2670	0.2627	0.2289
Average	0.2525	0.2585	0.2563	0.2460	0.2295

Table 3. The Pearson correlation coefficient (PCC) comparison of different features in interior design

	low-level visual features	high-level aesthetic features	rule-based features	human visual system features	combined features (ours)
Style 1	0.3961	0.4077	0.3758	0.3515	0.3459
Style 2	0.3438	0.3603	0.3771	0.3975	0.3521
Style 3	0.4036	0.3957	0.3699	0.3378	0.3322
Style 4	0.3611	0.4176	0.3520	0.3398	0.3443
Style 5	0.3579	0.3779	0.3878	0.3734	0.3345
Style 6	0.3792	0.3972	0.3823	0.3673	0.3514
Average	0.3736	0.3927	0.3742	0.3612	0.3434

the results in Table 1, 2, and 3, it can be found that the combined features is much better than merely using one type of features.

4. CONCLUSION

Now, the interior design has become an essential part in modern interior space, such as personal house, department, and flat. The aesthetic evaluation plays an important role in the modern interior design. With development of derivative digital devices, a large number of digital images have been emerged. This paper utilizes machine learning and computer vision to implement an intelligent aesthetic evaluation framework for interior design evaluation. First, the interior design images are collected by using digit camera. Second, the collected interior design images are represented as the combination of associated low-level visual features, high-level aesthetic features, rule-based features and human visual system features which are extracted from the images. Third, the extracted features are used to learn a regression model, such as support vector machine. The proposed interior design evaluation scheme is verified on the collected images from designers through digital camera.

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